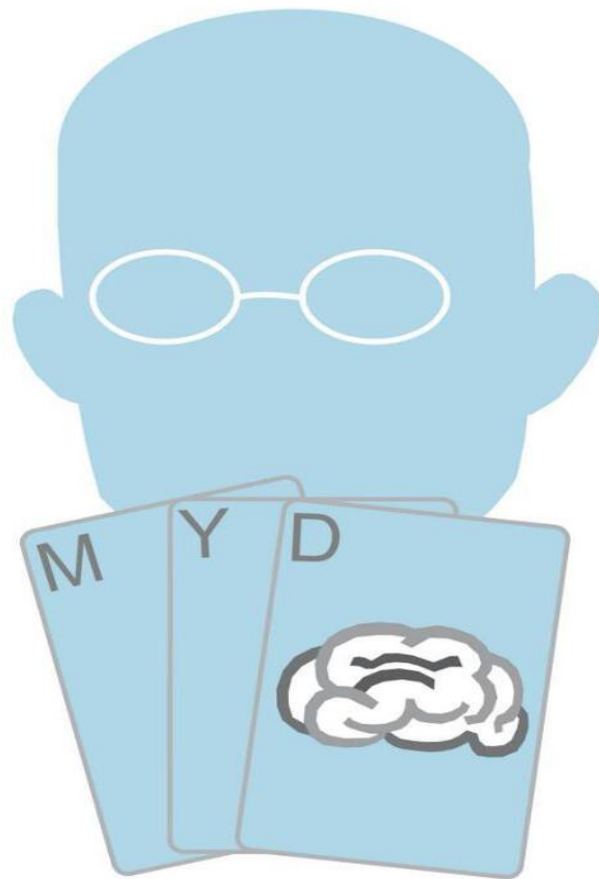


40 PARADOXES IN LOGIC, PROBABILITY, AND GAME THEORY



PRESH TALWALKAR

40 Paradoxes in Logic, Probability, and Game Theory

Presh Talwalkar

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About The Author

Presh Talwalkar studied Economics and Mathematics at Stanford University. His site *Mind Your Decisions* has blog posts and original videos about math that have been viewed millions of times.

Books By Presh Talwalkar

The Joy of Game Theory: An Introduction to Strategic Thinking .

Game Theory is the study of interactive decision-making, situations where the choice of each person influences the outcome for the group. This book is an innovative approach to game theory that explains strategic games and shows how you can make better decisions by changing the game.

Math Puzzles Volume 1: Classic Riddles And Brain Teasers In Counting, Geometry, Probability, And Game Theory . This book contains 70 interesting brain-teasers.

Math Puzzles Volume 2: More Riddles And Brain Teasers In Counting, Geometry, Probability, And Game Theory . This is a follow-up puzzle book with more delightful problems.

Math Puzzles Volume 3: Even More Riddles And Brain Teasers In Geometry, Logic, Number Theory, And Probability . This is the third in the series with 70 more problems.

But I only got the soup! This fun book discusses the mathematics of splitting the bill fairly.

40 Paradoxes in Logic, Probability, and Game Theory . Is it ever logically correct to ask “May I disturb you?” How can a football team be ranked 6th or worse in several polls, but end up as 5th overall when the

polls are averaged? These are a few of the thought-provoking paradoxes covered in the book.

Multiply By Lines . It is possible to multiply large numbers simply by drawing lines and counting intersections. Some people call it “how the Japanese multiply” or “Chinese stick multiplication.” This book is a reference guide for how to do the method and why it works.

The Best Mental Math Tricks . Can you multiply 97 by 96 in your head? Or can you figure out the day of the week when you are given a date? This book is a collection of methods that will help you solve math problems in your head and make you look like a genius.

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Table of Contents

[Introduction](#)

[LOGICAL PARADOXES](#)

- 1) [Knowing Without Knowing](#)
- 2) [The Case of the Missing Dollar](#)
- 3) [Lowering Both Averages](#)
- 4) [What Exactly is a Pile of Sand?](#)
- 5) [The Missing Square](#)
- 6) [“Proofs” that \$1 = 2\$](#)
- 7) [The Least Interesting Number](#)
- 8) [Selected Self-Referential Paradoxes](#)
- 9) [The Debt that Must be Paid?](#)
- 10) [The Paradox of the Question](#)
- 11) [Russell's Set of All Sets](#)
- 12) [A Paradox of Motion](#)
- 13) [How Can a Part be Equal to the Whole?](#)
- 14) [Infinite Series Paradox](#)

[PROBABILITY PARADOXES](#)

- 15) [Birthday Paradox](#)
- 16) [False Positive Paradox](#)

- 17) [Simpson's Paradox](#)
- 18) [Bertrand's Box Paradox](#)
- 19) [Boy or Girl Paradox](#)
- 20) [The Monty Hall Problem](#)
- 21) [The Necktie Paradox](#)
- 22) [The Friendship Paradox](#)
- 23) [St. Petersburg Paradox](#)
- 24) [Wild Card Poker Paradox](#)
- 25) [Gambler's Fallacy](#)
- 26) [Surprise Quiz Paradox](#)
- 27) [Two Envelopes](#)
- 28) [Siegel's Paradox of Exchange Rates](#)
- 29) [Allais Paradox](#)
- 30) [Ellsberg Paradox](#)
- 31) [Random Chord](#)

[GAME THEORY PARADOXES](#)

- 32) [You Can't Choose Wrong Paradox](#)
- 33) [Parrondo's Paradox](#)
- 34) [Centipede Game](#)
- 35) [Hypergame Paradox](#)

- 36) [The Wallet Paradox](#)
- 37) [Condorcet Paradox](#)
- 38) [Apportionment Paradox](#)
- 39) [Forward Induction](#)
- 40) [Braess Paradox](#)

[Conclusion](#)

[OceanofPDF.com](#)

Introduction

The word “paradox” comes from the Greek roots *para* (“against”) and *doxa* (“belief”). As used in this book, a paradox is a conclusion that is contradictory, counter-intuitive, or surprising.

This book is a collection of mostly logical and mathematical paradoxes that range from easy to understand and fun to resolve to bewilderingly hard to grasp and impossible to resolve.

The paradoxes are organized into three sections:

- (1) Logical paradoxes where reasoning leads to contradictory results.
- (2) Probability paradoxes that demonstrate the counter-intuitive nature of randomness.
- (3) Game theory paradoxes in which “irrational” strategies are surprisingly the “rational” option.

I have provided notes for each section that include citations and sources. It is not a traditional bibliography with consistent formatting, but rather a guide with enough information to be able to track down the original sources.

I love getting feedback. If you have a comment or suggestion, please email me presh@mindyourdecisions.com

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LOGICAL PARADOXES

This section is an assortment of mostly logical paradoxes, covering four main topics.

The opening paradoxes are about the difficulty in following logic precisely (like the *Paradox of Knowing without Knowing*). The next few are concepts that seem impossible to pin down with numbers (*What is the Least Interesting Number?*). Then there are a few *Self-Referential Paradoxes* (do you question the rule to “question everything”?). Finally, this section concludes with a few mathematical paradoxes involving infinity.

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1)Knowing Without Knowing

What distinguishes art from pornography? The difference is hard to define because even great artists have depicted nudity (Michelangelo's *David*) and used obscene language (Shakespeare¹ “peppered his plays with profanity”).

Furthermore, the question is important because art is protected free speech while pornography is not and can be censored.

The question was brought to the U.S. Supreme Court in the 1964 case *Jacobellis v. Ohio* . The manager of an arts theater, Nico Jacobellis, was fined by the state of Ohio for showing a French film considered obscene. Jacobellis contended the film was art, and as free speech, the state had no right to restrict it. Ultimately Jacobellis won the case, but there was a larger issue at hand. Would the Supreme Court explain the difference between art and pornography?

In short, no. The court opinions were divided and skirted the topic. Famously, Justice Potter Stewart dodged the issue by writing² , “I shall not today attempt further to define [pornography] ... and perhaps I could never succeed in intelligibly doing so. But I know it when I see it, and the motion picture involved in this case is not that.”

The ruling ended the case at hand, but it raised a logical dilemma. How was someone supposed to avoid pornography? Well, in order to view only art, one has to be able to identify pornography. But in order to identify pornography, one has to view it. The rule “I know it when I see it” does not settle the matters, but instead translates it into an unsolvable chicken and egg paradox.

The identification problem is not just limited to art and pornography. The paradox is that often one needs to know the answer to act, but it is only possible to learn the answer by acting. In other words, one needs to have the impossible talent of “knowing without knowing.”

Here are six more examples of the paradox.

(1) The Fifth Amendment of the United States Constitution provides that people have the right not to self-incriminate themselves in court. But how can a judge know in advance if the testimony would be self-incriminatory? The only way to know is by finding out what the circumstances are. But often it is the circumstances that the person wishes to avoid divulging. Judge Learned Hand noted the paradox³ : "Logically, indeed, [the witness] is boxed in a paradox, for he must prove the criminatory character of what it is his privilege to protect because it is criminatory."

(2) When is it correct to ask the question "May I disturb you?" The question is troublesome because the very act of asking it requires disturbing the listener. The question should really only be asked to someone who is okay being disturbed. But there is no way of knowing that in advance! The only way to find out if someone can be disturbed is by disturbing them, which essentially ruins the polite intention of the question.

(3) Companies advertise "You will love it" to get people to buy products, try new food items, and see movies. And yet, the company cannot necessarily know how people will react. The only way someone can actually say they love it is if they try it. And yet, in order for someone to want to try something new, they have to reasonably think they will like it, which is why the company needs to tell them "You will love it."

(4) In the TV show *The Simpsons* , Homer offers to tell his wife Marge a secret, but only if she promises in advance that she will not get mad. How can Marge know if she will get mad? She can only genuinely react to the news once she hears it; but she has to commit in advance in order to hear the news. Marge humorously replies that she *will* get mad, because she always does get mad when she has to make a promise that she will not. The idea of promising to react a certain way is a common joke in many TV shows and films⁴ .

(5) Imagine a picture of your private parts appears on the internet. You know the picture is a fake because it is anatomically wrong. But how can you prove it? You could release a genuine picture to remove any doubt, but that would obviously be counter-productive since exposing yourself publicly is precisely what you wish to avoid!

(6) The website Scribd is a document-sharing website. The website ran into trouble because users were uploading copyrighted material, such as leaking new Harry Potter books. In 2009 a lawsuit⁵ alleged Scribd was encouraging copyright infringement. Scribd replied that it did its best to filter out copyrighted material, and in fact, it was checking documents against an internal database. The funny part was this: the lawsuit alleged that the database was an act of copyright infringement too! The plaintiff's claim was that Scribd was supposed to filter out copyrighted material, but that it had to do so without a database to check against. How was Scribd supposed to know what is copyrighted without having copies of the copyrighted material?

The heart of the paradox is that knowledge is limited and you often cannot know the correct answer in advance. You have to do something and then later find out if it was justified.

In reality, people have to operate with limited information. In the examples above, here are some of the possible remedies.

(1) The Fifth Amendment privilege is often taken into context. The person might be able to provide non-relevant details to establish the testimony would be self-incriminatory, or the prosecutor might have reason to believe the person is lying and should be forced to testify on certain details.

(2) The question "May I disturb you?" is often not taken literally. The question is often a form of etiquette to mean, "I know you look busy, so let me check; are you really?" Someone who cannot be interrupted may simply remain silent and continue working as a signal that they are too busy to be disturbed. Someone who can be interrupted will feel obliged by the courtesy. Therefore, asking the question "May I disturb you?" is often better than simply approaching someone with no warning.

(3) Companies create previews of new products so people can "know" they will enjoy something before they have actually bought the product. There are a variety of preview methods. A technology company might do a live preview of their product at a conference. A company selling a subscription good might offer a free trial-period. A food company can distribute free samples. A final common example is known to any movie-goer. Movie

trailers are created so a viewer can get a sense of a film's plot without having to see the entire movie. All of these methods entice and persuade people so they will want a product before having the experience of buying and using it .

(4) Like the question “May I disturb you,” the request, “Promise me you will not...” is also not taken at face value. The requester wishes to signal the information is somehow embarrassing, hurtful, or confidential. The requester knows the person answering might not live up to the promise, but at least the stipulation was made explicit. Someone who breaks a promise one time might not be given a chance another time.

(5) People try to prove graphic pictures are fake by destroying the credibility of the person who leaked the picture. They also try to demonstrate their own morality and explain why they would never do such a thing.

(6) The lawsuit against Scribd for using a copyright filtering system was dropped⁶ . Publishers realized that Scribd does not have a magic way of filtering out copyrighted material.

In conclusion, while the paradox of “knowing without knowing” cannot be completely resolved, there are some practical workarounds that at least make the circumstances less difficult, and life a bit easier.

Notes

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2)The Case of the Missing Dollar

It was midnight and my consulting team was still in the office. We were working on an accounting case and we needed to audit our results using two equivalent derivations. But time and again when we crunched the numbers, we kept finding discrepancies. My boss joked, "It's like the story with the missing dollar." We had all heard this puzzle before, and yet in our sleep-deprived and mentally exhausted states, the riddle left us confused.

This is a classic accounting problem. Can you figure it out?

The story

Three gentlemen stay in a hotel room. The bill is \$30 so each of them pays \$10 for the room. The next day the hotel manager realizes he has overcharged them. The bill should actually be \$25. The hotel manager gives \$5 to the bellboy and tells him to return the money.

The bellboy thinks about how \$5 cannot be split evenly in three ways. Since the gentlemen don't know about the correction, he decides he can pocket \$2 and return \$1 to each person.

Now each of the three gentlemen has paid \$9 for his room, so that makes \$27. In addition, the bellboy has pocketed \$2. So here's the puzzle: the \$27 they paid and the \$2 the bellboy pocketed only adds up to \$29. But the room originally cost \$30. Where did the missing dollar go?

Try solving it before reading the answer.

The answer

Here's the reason: the story is a tale of misdirection because, of course, there is no missing dollar.

One method is to account from the original bill. Starting with the \$30 paid, \$3 made it to the gentlemen, lowering the total to \$27. Additionally, the bellboy pocketed another \$2, bringing the net balance to \$25. This exactly matches the amount on the revised bill.

The alternate method is account from the revised bill. The logic is exactly the same but the steps are reversed. The final bill was \$25, from which \$2 was pocketed by the bellboy, and also \$3 were refunded to the gentlemen. Adding the three brings the total to \$30, which was the amount of the original bill.

The apparent mystery is a result of confusing the two methods. The accounting in the story takes the \$27 paid and then incorrectly *adds* the \$2 the bellboy pocketed to arrive at \$29. This is just wrong. Either one adds the \$3 refund to the \$27 to arrive at \$30, or one subtracts the \$2 pocketed from \$27 to arrive at \$25.

The story demonstrates why it is important to carefully account for cash flows—it is easy to get confused!

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3)Lowering Both Averages

Consider two teams A and B where the average weight of the players on team A is 200 and on team B is 220 pounds.

What might happen to the average weight of each team if a player is transferred from team A to team B?

Intuition suggests that if the average weight of one team goes down, then the average weight of the other team should go up, or vice versa. After all, if a weight is transferred from one side of a weighing balance to another, one side would fall and the other would rise.

But that does not always happen when considering the *average* weight!

In fact, there are paradoxical switches that would make the average weight of both teams decrease (or equivalently make the average weight of both teams increase).

How is that possible?

Both averages decrease

Here is a mathematical example that illustrates the paradoxical change.

Suppose the players on each team A have weights {200, 200, 190, 210}, which is an average weight of 200 pounds.

Suppose players on team B have weights {220, 220, 190, 250}, which is an average weight of 220 pounds.

Now transfer the player who weighs 210 pounds from team A to B. What will the new averages be?

Team A now has players with weights {200, 200, 190}, which is an average weight of 196.67—clearly less than the average of 200 pounds they started with.

Team B, on the other hand, has players with weights {220, 220, 190, 250, 210}. The average weight of the team is now 218 pounds, which is also smaller than its starting average of 220.

So the average weight on both teams actually decreases!

Note that one can reverse the transaction to find an example of how the average weight on both teams can *increase*. (That is, transfer the player who weighs 210 back to team A. Now by reversing the calculations above one can see both teams have increased their average weights).

How is this possible?

The reason this is happening is pretty straightforward. The important factor is that the two averages were not the same to begin with.

When the 210 pound player was removed from team A, the team lost its heaviest player and so naturally that would lower the average weight.

At the same time, introducing the 210 pound player on team B was adding a player whose weight was less than the starting average of 220. Obviously this has the effect of lowering the average weight.

So the trick was this: the player that moved teams had a weight *in between the starting average values of the two teams*. That had the effect of moving both averages in the same direction—it decreased the average for the team that lost its above-average value, and it lowered the average for the team that got a sub-average value.

What this paradox teaches us

The counter-intuitive result that both averages could increase (or decrease) is known as the *Will Rogers phenomenon*. Will Rogers was an author and a humorist. During the 1930s Depression, some workers from Oklahoma moved to California in search of jobs. Rogers is quoted as saying¹, “When the Okies left Oklahoma and moved to California, they raised the average intelligence level in both states.” The statement is a joke to indicate that while the “Okies” (people from Oklahoma) were a group of below-average

intelligence people in Oklahoma, they were still smart enough to be classified as above-average in California. Will Rogers was born in Oklahoma.

The Will Rogers phenomenon is an important statistical concept in medicine. For instance, consider the above example re-framed with team A and team B being two different states. If a group of heavy people from state A migrated to an even heavier state B, statistics might bear out that the average weight in both states was decreasing. It could be easy to conclude that both states were getting “healthier.” In fact, the Will Rogers phenomenon demonstrates the improvement might be illusory; the change could just be a consequence of regrouping the data.

Notes

1. The “Okies” quote is attributed to Will Rogers. It appears in a 1985 medical paper by Dr. Alvan Feinstein, et. al. in the *New England Journal of Medicine*, “The Will Rogers Phenomenon — Stage Migration and New Diagnostic Techniques as a Source of Misleading Statistics for Survival in Cancer.” Feinstein proposed the term to point out potentially illusory progress in the treatment of cancer patients. The original paper: <http://www.nejm.org/doi/full/10.1056/NEJM198506203122504> . I learned about the original source from a 2009 paper by MP Sormani, Ph.D., *The Will Rogers phenomenon: the effect of different diagnostic criteria* . <http://www.ncbi.nlm.nih.gov/pubmed/20106348>

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4)What Exactly is a Pile of Sand?

A grain of sand is just a grain of sand. And just a few grains of sand are an insignificant collection of sand as well. But if enough grains of sand were present together, then eventually the group will be considered a pile of sand. The question is, at what point does the collection become a pile? No individual grain seems to create the mound, and yet eventually a pile of sand is created. Which grain is the pivotal one?

The Greeks called this the *Paradox of Sorites* (roughly translated as “paradox of the heap”). The heart of the paradox is that each individual grain of sand has a negligible contribution, but enough grains together will make a pile.

The paradox is a matter of vagueness. What exactly is a pile of sand? One could, for instance, do a survey of people to determine what amount of sand constitutes a pile. From there, one could quantify the subjective opinions and count how many grains are necessary for a pile. But then again, it seems hard to determine a specific standard. If the study concluded that 10,000 grains were needed to make a pile, it would be very likely a collection with one grain fewer, 9,999 grains, would also be considered a pile, so the paradox remains. It seems impossible to demarcate a lower bound because a pile of n grains and a pile of $n - 1$ grains are virtually indistinguishable.

The problem of the heap applies to any situation in which an individual component makes a seemingly negligible contribution but a large enough collection is significant. Here are 8 similar problems of vagueness.

(1) A single voter almost never sways a national election, but enough voters could change the outcome.

(2) Eating an extra calorie will not ruin a diet, but a collection of snacks can cause weight gain.

(3) A single molecule of air is insignificant, but enough of them can make a tornado or hurricane.

(4) A pinch of spice might not affect the flavor of a dish, but enough spice will .

(5) An extra credit point often has an insignificant impact on a class grade, but earning sufficient marks is the difference between an A and B grade.

(6) A butcher will happily offer a slice or two of meat for a sample, but if he keeps giving slices away he will go bankrupt.

(7) Losing a single strand of hair means nothing, but losing enough hair makes one go bald.

(8) Having an extra sip of beer won't affect one's ability to drive, but at some point imbibing alcohol does impair coordination, making driving unsafe.

Resolving the paradox

In the above examples, each individual action has a negligible impact, but a collection of actions is a slippery slope to a stark outcome.

One possible remedy is to create a demarcation point, even if that point is somewhat arbitrary. For instance, here are some standards commonly used in the eight examples.

(1) A bloc of 10,000 voters in a swing state might be considered influential.

(2) A diet might demand a specific calorie limit to counter the temptation of small snacks.

(3) Meteorologists consider certain wind speeds (say 20 mph and higher) to be dangerous.

(4) A certain amount of spice, like 1/8 of a teaspoon, could be considered a pinch.

(5) Teachers set specific percentage breakdowns for grades.

(6) Butchers will get fed up if a patron asks for more than 5 samples (or a certain number) without buying anything.

(7) People ask around if their hair is “thinning” and realize they are going bald .

(8) The government sets a specific alcohol tolerance level for drivers—or perhaps they might even say there is zero tolerance for any amount.

What this paradox teaches us

Specific standards are easy to mock. After all, why does a diet specify exactly 1,600 calories, or why does the U.S. government think that a driver is safe at a blood alcohol content of 0.079, just below the legal limit of 0.08, but suddenly the driver is dangerous at 0.081? And why does getting 90 percent mean an A grade but getting 89 percent mean a B grade? Did scoring just one percentage point less mean the student is not up to par?

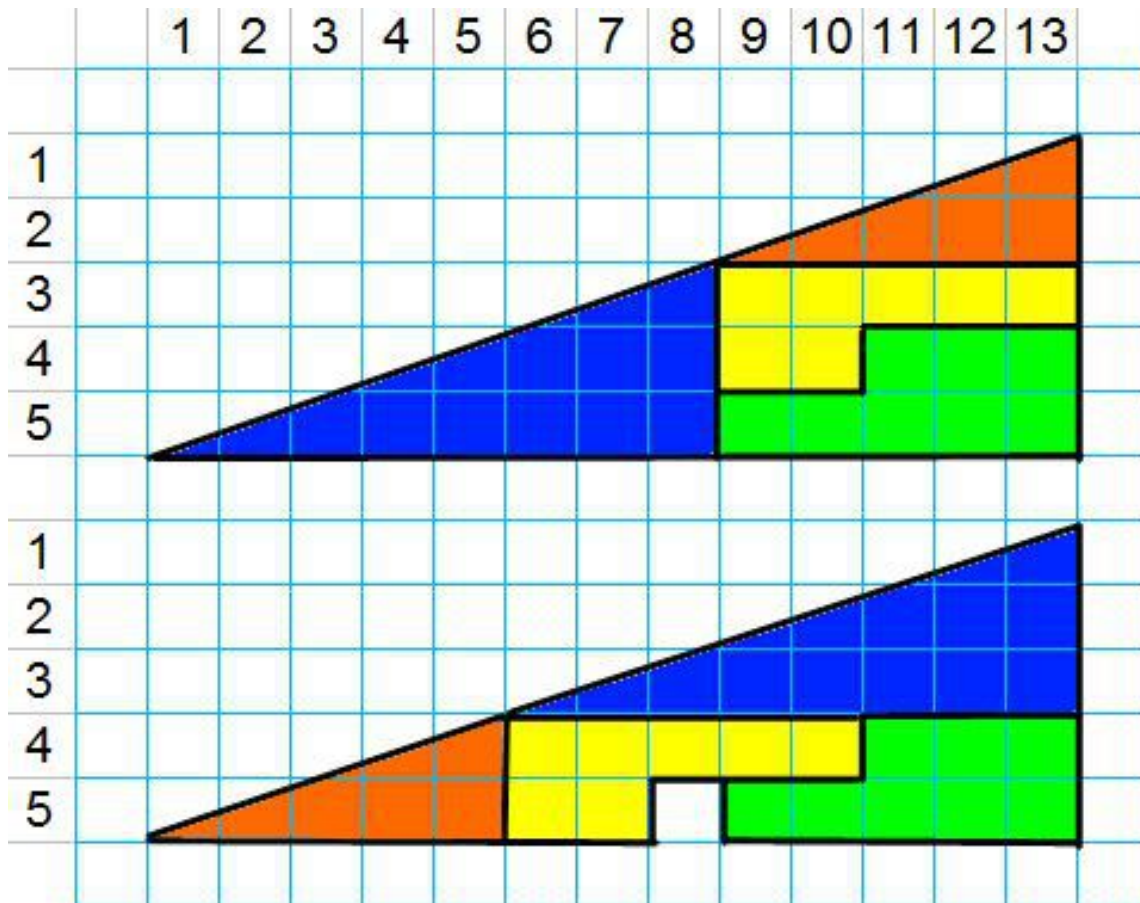
It is easy to ridicule specific standards. But this paradox demonstrates exactly why these standards might be needed: without a clear standard, there can be no objective way of distinguishing one outcome from another.

So we are left to choose between an objective standard or remain in a vague situation where everything can be argued. It appears there is a good reason most organizations opt for the clear standards, even if they might suffer from being arbitrary.

5)The Missing Square

This is a visual paradox. Below are two figures decomposed into smaller shapes.

Why is there a “missing” square in the second figure?



A mathematical excursion

Remember the Fibonacci numbers? The sequence starts out with the numbers 1 and 1, and then each subsequent number is the sum of the previous two numbers.

The next numbers are $2 = 1 + 1$, then $3 = 1 + 2$, then $5 = 2 + 3$, and so on. The sequence is 1, 1, 2, 3, 5, 8, 13, and so on.

There are many interesting properties about the Fibonacci sequence. The one that is relevant to this paradox is a special relation between the numbers.

First a bit of notation. Write F_1 for the first Fibonacci number, F_2 for the second, and generally F_n for the n^{th} term. There is a special equation known as Cassini's formula that relates three sequential terms.

The formula is: $F_{n+1} F_{n-1} - (F_n)^2 = (-1)^n$, for $n > 0$

The formula states that the square of a Fibonacci number is equal to the product of the preceding and succeeding terms, plus or minus a factor of 1.

Why does this formula matter for this problem?

Take a look at the shapes in the figure. All of the shapes have length or widths of 2, 5, 8, or 13, which are Fibonacci numbers. The trick is based on Cassini's formula that $(13)(5) - (8)(8) = 1$

That extra “1” is essentially where the missing square comes from.

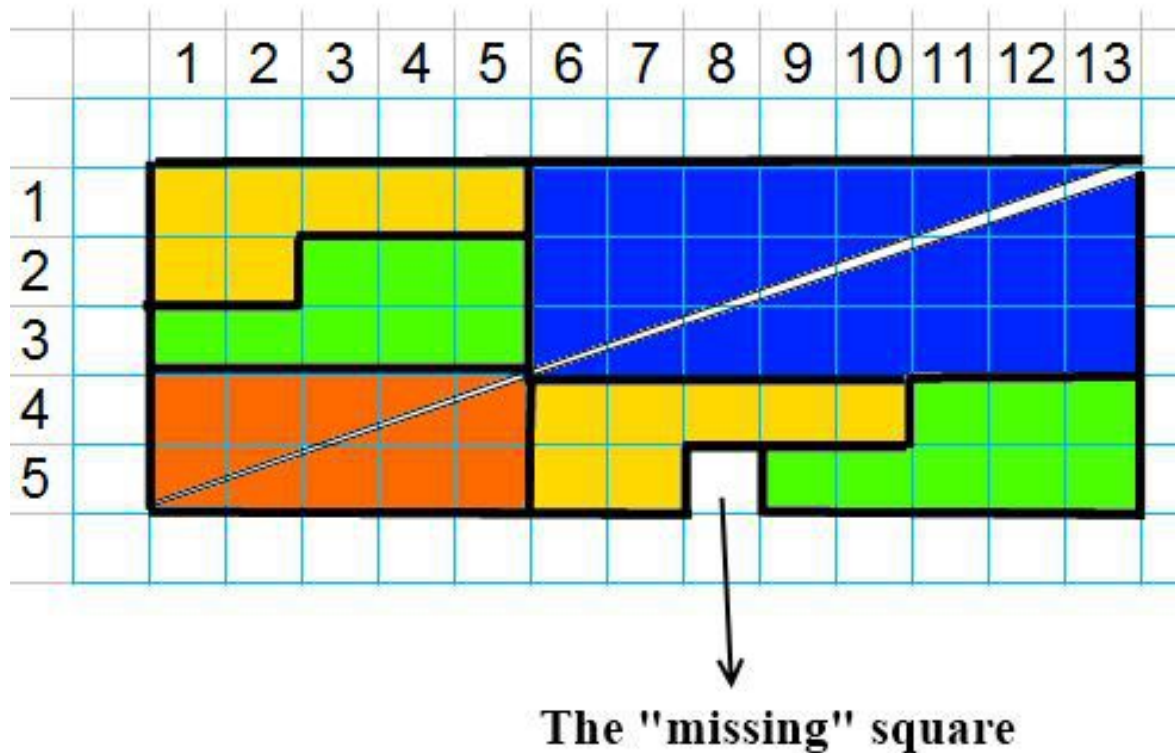
Here is a bit more of the explanation. The top triangle is a true triangle with a height of 5 and a length of 13, so its area is half of $(13)(5)$.

The bottom triangle is not really a triangle. It is actually close in area to half of an 8 by 8 square, with one square removed. The problem utilizes the fact that $(8)(8)$ is one less than $(13)(5)$, which is what makes the optical illusion convincing and difficult to detect.

In the following figure, the two shapes are juxtaposed. The upper left shape looks like a right triangle. The bottom shape has the missing square.

In between the two figures there is a small sliver of area (it's very small so it's hard to show graphically). This area which is shaved is the reason for the “missing” square.

**The two triangles are shown next to each other
The bottom "triangle" is not really a triangle
The missing square is just an optical illusion**



In conclusion, the missing square is not really missing at all. It is just the circumstances create an illusion of a missing quantity.

And a concluding bit of trivia: the Indian mathematician Pingala came upon the same sequence of numbers 1,200 years before Fibonacci in the study of Sanskrit poetry.

6)“Proofs” that $1 = 2$

These are not true paradoxes because the conclusions are clearly false. And yet these types of problems are interesting because they take a little bit of thought to figure out the mistake.

Often, such a problem is presented as a test to students learning algebra. Take a moment to read the following proof and do a self-test.

“Proof” that $1 = 2$

Each of the steps seems to be correct, but again, the conclusion is wrong. Where is the mistake?

$a = b$	Assumption
$a^2 = ab$	Multiply both sides by a
$a^2 - b^2 = ab - b^2$	Subtract b^2 from both sides
$(a + b)(a - b) = b(a - b)$	Factor both sides
$a + b = b$	Divide by $(a - b)$
$a + a = a$	Substitute that $a = b$
$2a = a$	Simplify
$2 = 1$	Divide by a

After solving that problem, tackle the next “proof” that demonstrates how 2 equals 3.

“Proof” that $2 = 3$

$4 - 10 = 9 - 15$	
$4 - 10 + 25/4 = 9 - 15 + 25/4$	Add 25/4 to both sides
$(2 - 5/2)^2 = (3 - 5/2)^2$	Factor
$(2 - 5/2) = (3 - 5/2)$	Square root
$2 = 3$	Add 5/2 both sides

Where is the mistake?

Why the proofs are wrong

In the first example, the proof breaks down at the step of dividing both sides by the term $(a - b)$. The proof begins with the assumption $a = b$, so the step of dividing by $(a - b)$ is the equivalent of dividing both sides of the equation by 0, which is a prohibited operation. The proof essentially has reasoning along the lines of “Since $2(0) = 1(0)$, which is true, dividing both sides by 0 leads to the conclusion that $2 = 1$.” The paradox is resolved by removing the illegal operation.

The second example¹ is wrong for a different reason. The mistake happens at the step of taking the square root of both sides. Before taking the square root, there is the valid expression that $(2 - 5/2)^2 = (3 - 5/2)^2$, which simplifies to $(-0.5)^2 = (0.5)^2$. After taking the square root of both sides, it is not proper to conclude the two quantities are equal. The correct conclusion is that either the two quantities are equal OR they are the opposites of the other (which is the case), and that resolves the apparent paradox.

While these “proofs” pose no problem to mathematics, they are interesting examples of why mathematical rules are important.

Notes

1. The example $2 = 3$ is adapted from a paradox found on this website:
http://www.clockstone.com/enigma/algebra_paradoxon_2.htm

7)The Least Interesting Number

What's the least interesting positive number in the world?

Well, it can't be 1 because 1 is a very important number: it is the smallest positive number and it is the multiplicative identity. The answer clearly can't be 2 since 2 is the smallest positive even number, plus it is the only even number that is a prime number. Then again, the answer also can't be 3 since 3 is the smallest odd prime number...

It would be easy to describe why the numbers from 1 to 100 are interesting. But there's really no point to continuing this process since there can't be a logical answer anyway!

Here is the reason why. One possible solution might be to conduct an election. One could nominate candidates for the least interesting number and then take a survey.

Suppose an online voting contest is conducted and that some number is called the least interesting number. Now that there is a winner, does that answer the question?

Not in the least. You see, any number that wins has the property that people considered the number to be the least interesting. Thus, this number won a contest, which is an accolade none of the other numbers have. That alone makes the winner at least a little bit interesting.

A possible solution is to disqualify the highest vote-getting number and crown the runner-up as the least interesting number.

But then again, if the runner-up gets the title, that again would make for a nice accolade, essentially negating the number's uninteresting nature. It must be the third place winner that should be considered...

You can see how this logic can cascade to eliminate the remaining candidates one-by-one.

Therefore, no number could possibly be the least interesting. And they say math is boring ?

A potential interview question?

Imagine a technical interviewer asks you the following question. What's the least interesting thing about you?

How do you answer?

Many people might reply by explaining how “normal” they are in terms of physical appearance (“I’m average height”) or maybe in terms of a food preference (“I enjoy McDonald’s”).

In fact, the question is a sort of trick along the lines of the least interesting number paradox.

Imagine you could rank your attributes in terms of most to least interesting. Well, once you’ve come up with the least interesting aspect, one could argue that aspect itself must be interesting because it is the least interesting!

On a positive note, this logic suggests there is really nothing uninteresting about you at all. On the downside, such self-importance might explain the billions of boring updates on Facebook and Twitter.

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8)Selected Self-Referential Paradoxes

Consider the rule: “All rules have exceptions.”

The rule seems fine. But the problem is if that rule is completely true, then that rule should also have an exception—that is, there should exist some rule with no exceptions. This conclusion contradicts the original rule.

The problem is the rule specifies a statement about “all rules.” In other words, the rule is referencing itself, or is *self-referential* .

A potential resolution is to modify the statement to be: “All rules have exceptions, except this one.” Now there is seemingly no trouble with the rule, unless you think very carefully about it.

Parents are well versed with the “rules have exceptions” paradox. A parent might tell a child never to lie, only to be caught lying the very same day. So to cover, the parent replies, “Do as I say, not as I do.” That sentence itself is not a paradox, but it is perhaps the more puzzling concept to understand.

There are many variations on the idea of a self-referential paradox; below is a sampling of them.

The Barber Paradox

This is one of the famous paradoxes by the logician Bertrand Russell.

Consider a barber who shaves everyone in the town except the people who shave themselves. The question is, who shaves the barber?

If the barber does in fact shave himself, that is not really allowed: he is only supposed to shave people who do not shave themselves.

If the barber does not shave himself, then he has classified himself as someone that he should be shaving.

Evidently the barber should and should not shave himself. What should he do ?

This seems to only be a logical problem and not a practical one, as no barber in the history of the world seems to have ever reported this dilemma.

The Liar Paradox

Try this one out for size: “This statement is false.” Is that a true or a false statement?

Note that if the statement is false, that means in fact it was a false statement, and hence the sentence was factually true.

On the other hand, if the statement is factually true, then that means the statement was wrong about being false. In other words, it was a false statement, which again is a contradiction to the premise.

In science fiction shows, the liar paradox always manages to foil evil logical villains or robotic computers who are uncomfortable with the contradictory logic. Humans apparently have no trouble with this sort of doublethink.

The Card Paradox

On a postcard there are written statements on each side.

Side A has the sentence: “The statement on the other side is true.” Side B has the sentence: “The statement on the other side is false.”

Which statement is true, if any?

One can work through the logic to see neither statement can be true without leading to a contradictory conclusion.

Pinocchio Paradox

This is a variation of the Liar Paradox that was created by an 11-year old who is the daughter of an Australian logician¹.

Pinocchio is a fabled wooden toy that becomes alive as a boy, with one special characteristic: his nose grows whenever he tells a lie. In the story,

Pinocchio is caught telling a lie when he tries to conceal money, and eventually his nose keeps growing as he fibs .

Pinocchio's nose is a physical lie detector and that's good enough for story time. But what if Pinocchio were a logician or a computer programmer? Imagine the fun he could have if he tested his nose out.

Let's say he makes the innocuous statement along the lines of a Liar Paradox, "I am lying." Now, if Pinocchio were truly lying, that would mean his statement is false, and he is actually telling the truth. But that would mean the statement was accurately described as a lie, so he was telling the truth. Would his nose grow or not?

There is an even cleverer thing he could say. Imagine that Pinocchio says, "My nose will grow." If he is telling the truth, then his nose should not grow, which will on net make his statement false and make him a liar. If he is lying, then his nose will grow in response. But that would mean his statement was factual and so he was not in fact lying! It seems Pinocchio can trick his nose if he really wanted to.

"I Know Nothing" Paradox

This is a paradox attributed to Socrates.

If you state you know nothing, then you are claiming to be ignorant of any factual statements. If that were true, then your statement would be true and so you at least know that!

At best, you could state your ignorance as, "I know nothing, except this statement."

Frightened Inmate Paradox

This is a scenario that takes place in one of my favorite TV shows, *Arrested Development* . In the episode "Marta Complex," the character Tobias is auditioning for a role with the description "Frightened inmate #2."

Tobias, a fledgling actor, is very worried about landing this minor role. In the plot, the narrator describes the paradox: “Tobias arrived at his audition for Frightened Inmate #2. The competition frightened Tobias, which he felt he could use in his performance. Unfortunately, this made him more confident, which frightened him again.” Poor Tobias .

This situation is closely related to the *Paradox of Fiction* . For fiction to be emotionally moving, the audience has to become attached to the characters and somehow think they are real. And yet, deep down, the story is known to be fictional. So how can an audience ever become attached to things that are known not to be real?

Clearly they can, as the tears after a heart-throbbing drama or the screams during a horror film attest. It might be because fictional stories lead to thoughts and reflection about things that do exist and are genuinely sad or scary.

No Comparison

Have you ever heard someone say, “Oh, there is no comparison to this drink/cuisine/car/clothing brand?”

Now, if there were truly nothing to compare, then there would be no reason to make the statement “there is no comparison.” On the other hand, the fact there are things you can compare means the statement “there is no comparison” is false!

The expression of course means the item in question is of excellent quality and that one should not compare it to inferior products. But if that is the case, why not just say that?

Paradox of Tolerance

Tolerance is the idea that you can “tolerate” and respect people with other viewpoints. Tolerance can be a great virtue in that it means people of different races, religions, and opinions can peacefully get along.

But let's say that someone is bigoted and truly holds intolerant views. Will that person be tolerated by those who profess tolerance?

Absolutely not!

You see, tolerance is something that is intolerant of intolerance. You can't be tolerant towards everything, otherwise that means it is fine for people to be bigoted.

The paradox is somewhat captured in the quote, “I have seen great intolerance shown in support of tolerance,” by Samuel Taylor Coleridge .

This is a concept that was mocked in a *South Park* episode “The Death Camp of Tolerance” where the kids are sent to a camp to learn the importance of tolerance, only to learn that means they cannot tolerate intolerance.

Omnipotence Paradox

If a being is all-powerful, can it make a rock so big that it cannot lift it? If it can make such a rock, then it is clearly not all-powerful and hence not omnipotent. If it cannot make such a rock, then that again demonstrates a limitation, which again means the being is not omnipotent.

The paradox has been discussed for a long time because of its theological significance. Some people point out the flaw is in assuming such an omnipotent being exists, while others are content to clarify how an omnipotent being might be outside of humanity's logic system.

A similar paradox is the “irresistible force paradox” which wonders what would happen between an immovable object and an unstoppable force. This is often the philosophical dilemma faced between a villain and a super-hero, as neither can evidently achieve greatness without the other. This very paradox is stated by the Joker in the film *The Dark Knight*. Generally something gives, like the hero tricking the enemy, or maybe sacrificing himself to save the world. As for the resolution to the physical problem, perhaps the main issue is assuming that such impossible objects exist.

Notes

1. Peter Eldridge-Smith and Veronique Eldridge-Smith. "The Pinocchio Paradox." *Analysis* (2010) 70(2): 212-215 first published online January 13, 2010 doi:10.1093/analys/anp173

<http://analysis.oxfordjournals.org/content/70/2/212>

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9)The Debt that Must be Paid ?

The comedian Stephen Wright joked he once bought a humidifier and a de-humidifier for the purpose of setting them both in the same room to “fight it out.”

A similar conflict is present in the following paradox. Consider a student in the following situation: “My dad paid for my law school, offering that I could repay him after I won my first court case. After I took the loan and paid my tuition, he sued me for the entire amount.”

By a careful analysis of logic, the seemingly innocent loan sets up a no-win situation for the debtor, who faces two potential outcomes.

(A) Chances are the outlandish lawsuit would be thrown out, meaning the student would have won his first case. But in that case, he would be required to pay up per the original agreement.

(B) In the other case, if the dad somehow did win, that would mean the student has a legal obligation to pay the money back.

Then again, the student could equally argue the case means he does not have to pay. He could argue the following two circumstances.

(A) If he won the case, that would mean he has no legal obligation to pay.

(B) If he lost, then that would mean he still has not won a case, and hence, he should not have to pay!

Such a story is said to have happened in Greece between the philosopher Protagoras who offered a loan to his student Euathlus. (Protagoras was either greedy or Euathlus was a deadbeat, depending on which side you believe). Apparently the court was so confused that it adjourned for 100 years¹ .

It might be a little bit arrogant, therefore, to speculate on what the correct answer should be. But let me do that anyway. In today's legal setting, the court ruling would probably take precedence. If the court decided the

student did or did not have to pay, then the previous agreement could be annulled by the court's ruling. After all, the whole point of going to court is to try to settle the matter.

Notes

1. Suber, Peter. *The Paradox of Self-Amendment*. Available online.
<http://legacy.earlham.edu/~peters/writing/psa/sec20.htm>

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10)The Paradox of the Question

This is a delightful paradox written about by Ned Markosian in 1997¹ .

One day, an angel visits you and presents an offer of a lifetime. The angel offers to answer *any* question you wish to ask. You are allowed to ask about anything, but the angel emphasizes you can only ask a single question.

You realize the importance of getting the question right, and you ask around for advice. You post your situation on your blog, you tweet about it, you make a group on Facebook, and you email every single professor and smart person you know. Your post manages to blow up and suddenly everyone on Reddit is offering suggestions about the best possible question.

Some people ask practical questions (“What is the best diet to eat?”); some people make the question political (“How can we reduce gun deaths in America?”); and a few people ask questions for global good (“Can you tell me the cure for all diseases?”).

Some people think even grander, wondering to find out what the best possible question would be. But asking that question would be stupid, as it would waste the only question you are allowed to ask.

Ultimately, a mathematician proposes a neat idea that catches your attention. The person realized the logical trap of asking about the best question, and so the mathematician came up with a clever request to capture both the question and answer. The mathematician's question is: “Can you tell me the ordered pair (x, y) , where x = the best possible question to ask, and y = the answer to x .”

You think, this is brilliant! The question is just one question, and it sneakily asks for an ordered pair to learn about a question and answer.

You submit this idea on Reddit and it gets tremendously upvoted beyond every other option. Many professors, CEOs, and several Nobel Laureates agree this is the best possible question to ask.

Confident that you have done an exhaustive search, you finally get the nerve to ask the angel. You set up a live stream on your blog and billions of people tune in and anxiously await the answer.

After a moment, the angel offers the reply: “The answer to your question is that the best possible question x = the question that you asked me, and the answer to that question y = the response that I am giving to you right now.”

The angel disappears before your eyes, leaving you and the entire world stunned in silence. The question seemed to be perfect, and yet the angel's answer was so worthless that it provided no useful information to the world. You regret that you didn't just ask about the ending to *The Sopranos* . Even if you had asked a trivial question like, “What does $6 \div 3(1+2)$ equal?” you could have at least ended that Facebook debate of whether the answer is 9 or 1.

And yet, by asking the smartest question in the world—as the angel confirmed you did—you ended up learning nothing. The paradox is, how can asking the best question be wrong?

There are a few ways to resolve the paradox. One issue is presuming such an angel exists and would offer to answer a question. Remember this is a story so there is no real relevance. Another version of the story could have resolved with you learning the actual answer but then sleeping on it and forgetting it. The point is, it can be a mistake to over-analyze a hypothetical situation.

Another issue is the angel did not actually provide the best answer, since the best answer should benefit humanity in some way. Angels are supposed to do good things, not force people into logical traps.

A final issue is the question asked really was two questions. If you could ask for a pair (question, answer), then why not extend the logic to ask even more questions? Perhaps the question could be: “Can you provide me with the ordered n -tuple with the answers to how to end world hunger, cure all diseases, etc.”

Ultimately the problem is one of presupposing the scenario. It would be nice to have a supernatural angel answer an important question. For now, we have to settle with the solutions that people find .

Notes

1. Markosian, Ned. "Paradox of the Question." *Analysis* 57 (1997), pp. 95-97. <http://myweb.facstaff.wwu.edu/nmarkos/Papers/PQ.pdf>

The format of the question (x, y) is borrowed from Sider, Theodore. "On the Paradox of the Question." *Analysis* 57 (1997): 97–101.
http://tedsider.org/papers/paradox_of_question.pdf

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11) Russell's Set of All Sets

This is a more technical paradox than the ones before, but it is an interesting one from set theory. A mathematical *set* is a collection of elements. For instance, the set $S = \{a, b, c\}$ has three members, the letters a , b and c . The set S itself is an object too, and the set S is not a member of itself.

Consider another set T defined as “the set of everything not in S .” The set T will be very large, as it contains everything in the universe that is not the three letters a , b , and c . And one thing that T will contain is the set S . In other words, the set S is a member of T .

Going one step further, set T is not a member of the set S . Therefore, the set T is a member of itself. The whole point of this setup is to realize that some sets are members of themselves (like T) while others are not (like S).

Now comes the interesting part.

The set of all sets

The logician Bertrand Russell defined a set R as follows: “Let R be the set of all sets that are not members of themselves.”

The question is, is R a member of itself? Consider the logical possibilities:

(1) If R is not a member of itself, then R is a set that is not a member of itself. But by its definition, that necessarily means R must be included in R , a contradictory conclusion.

(2) If R is a member of itself, then R has the property that it is not a member of itself. This again contradicts the premise.

The logic breaks down because R is referring to itself. This paradox is a more general version of the *Barber Paradox* discussed previously.

One practical resolution—employed by standard set theory today—is that such sets R are not properly defined and so in effect they do not exist. If prohibiting certain sets sounds like a cop out, remember that other areas of

math also preclude troublesome operations, such as the prohibition on dividing by zero. Developing the right rules is an important step to creating sensible mathematical conclusions.

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12)A Paradox of Motion

This is one of the many delightful paradoxes of motion and time attributed to the ancient Greek Zeno.

Clearly motion exists as people experience the joy of movement. Yet Zeno made a clever argument of why motion should never happen.

Here is the argument. Imagine you want to run 1 meter. To reach the goal, you have to move half of that distance, which is $\frac{1}{2}$ of a meter. Once at that point, you still have to move half of the remaining half-meter, which is $\frac{1}{4}$ of a meter. The logic can continue: at any point, you still have to move half of the remaining distance to reach the goal, so one then has to travel $\frac{1}{8}$ of a meter, $\frac{1}{16}$ of a meter, and so on. In other words, at any point, there is still space between you and the endpoint equal to half the remaining distance. Since there will always be some space, you will never reach the endpoint. Hence, motion is impossible.

Zeno's argument is baffling because it is logically reasoned and yet readily contradicted by physical experience. You can easily disprove the conclusion by standing up and walking one meter.

But the argument cannot be dismissed simply because its conclusion is wrong. One has to figure out the flaw in the logic.

The paradox can be resolved mathematically in an elegant proof using infinite series. Zeno's argument is that at each stage there is some halfway point that needs to be reached, and so there is always some “space” between the current point and the endpoint. Perhaps it is a mistake to focus on the “space” argument because it makes it seem there must always be a positive distance remaining. Mathematically, one could argue the spaces keep diminishing in size and get infinitesimally small, so small that eventually there is no space left.

But perhaps a clearer resolution is to focus on the distances traveled. Since you are traveling $\frac{1}{2}$ of a meter, then $\frac{1}{4}$ of a meter, and so on, we can equivalently add up the amount of distance traveled. If that distance adds up to 1 meter, then that means you can travel the distance without problem .

The infinite series of distances traveled is the following equation.

$$(I.) S = 1/2 + 1/4 + 1/8 + \dots$$

The hope is the series adds up to 1 meter. What is the value of S ?

The trick is to manipulate the sum algebraically by halving both sides. Doing this yields the following equation.

$$(II.) \frac{1}{2} S = 1/4 + 1/8 + 1/16 + \dots$$

Note that the infinite series in equation (II.) is precisely the same series in (I.) except that it does not include the $1/2$ term. Therefore, if the equation in (II.) is subtracted from (I.), every term in the infinite series after $1/2$ gets canceled out.

Thus, the subtraction of the two equations leads to the result $\frac{1}{2} S = \frac{1}{2}$. The conclusion is that $S = 1$, which means the set of distances does add up to 1 meter.

Zeno's paradox is an example of how an infinite series can be used to satisfactorily resolve a logical argument with practical experience. This is not always the case as illustrated in the next two paradoxes.

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13)How Can a Part be Equal to the Whole?

Galileo is attributed with this paradox about infinity. Now clearly the set of positive numbers 1, 2, 3, ... and so on is infinite. For any number n , there is a larger number $n + 1$, meaning the sequence of numbers never runs out.

The question is, how big is the set of even numbers by comparison?

The intuitive answer is that the even numbers must be smaller. That is because they are a subset of all numbers, and a part should be smaller than the whole.

Galileo was someone who thought this was not correct. In fact, a careful mathematical argument can show the set of even numbers is *equal* in size to the set of all positive numbers.

One has to take a step back in order to make sense of this statement. What exactly does it mean to say that two sets have the same size?

In mathematics, the concept of size (or *cardinality*) is a process of matching. Roughly speaking, the size of a set is determined by how it matches up with the set of positive numbers (or if it cannot match up, then it would be larger and uncountable).

A single person has a size of 1 because it can be matched to that number in a correspondence: single person $\leftrightarrow 1$

Similarly, a pair of people will have size 2 because the pair can be matched to the set of numbers $\{1, 2\}$ as follows: person A $\leftrightarrow 1$, person B $\leftrightarrow 2$.

So when a set is said to be a particular size, it means the set can be matched in the process described.

Now, return to the question: how big is the set of even numbers? Using the matching principle, the even number 2 can be matched to 1, the even number 4 can be matched to 2, and in general, the even number $2n$ can be matched with the number n . This can be written out as: $1 \leftrightarrow 2, 2 \leftrightarrow 4, 3 \leftrightarrow 6, \dots, n \leftrightarrow 2n, \dots$

The list will continue forever, but the important detail is this: the two sets are matched up perfectly. Mathematically that means the two sets have the same size, or in technical terms, the same *cardinality* .

This example shows that infinity is a strange beast: a portion of the whole might still be of equal size to the whole.

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14) Infinite Series Paradox

This problem¹ is a bit more mathematical than the previous ones in this section. But it can be solved by with good reasoning skills and a little recollection of high school algebra.

See if you can figure it out.

Begin by defining the infinite series where the n^{th} term is the fraction $1/n$, as follows

$$(I.) S = 1 + 1/2 + 1/3 + 1/4 + 1/5 + \dots$$

The next step is to take half of the series. That is, multiply both sides of the equation by one-half. The end result is the following series where every n^{th} term is the fraction $1/(2n)$.

$$(II.) \frac{1}{2} S = 1/2 + 1/4 + 1/6 + 1/8 + 1/10 + \dots$$

The next step is to subtract the series in (II.) from the series in (I.). This can be accomplished term by term. Since the second series has every even fraction $1/(2n)$, and the first series has every fraction $1/n$, subtracting (II.) from (I.) results in a series with only the odd fractions.

$$(III.) \frac{1}{2} S = 1 + 1/3 + 1/5 + 1/7 + 1/9 + \dots$$

Now for the final step. Both the series in (II.) and (III.) are different expressions for $\frac{1}{2} S$. Subtracting (II.) from (III.) yields the following nested series.

$$(IV.) \frac{1}{2} S - \frac{1}{2} S = 0 = (1 - 1/2) + (1/3 - 1/4) + (1/5 - 1/6) + \dots$$

Since we are subtracting $\frac{1}{2} S$ from itself, the result should be zero. But each term in the nested series is clearly positive, and therefore the sum of the series is greater than zero.

How is this possible? In other words, what went wrong?

The harmonic series

The series S where the n^{th} term is $1/n$ is also known as the harmonic series. What is its sum?

A good approach to problems like this is to work numerically. That is, add up some of the terms and see if there might be a pattern. The first ten terms (from 1 to $1/10$) sum to about 2.92. The first 100 terms sum to about 5.2. The first 1,000 terms sum to about 7.5. Adding up a lot of terms, say the first 10^{43} terms, leads to a sum² of just over 100. At this point, one might conclude the series sums to about 100. The reason is that the remaining terms are incredibly small fractions that would not seem to affect the sum.

Remarkably, that intuition turns out to be false. It can be proven the harmonic series does not converge. Even though the sum grows very, very slowly, like the Energizer Bunny, it keeps on going and going and going.

Here is a proof for why the harmonic series does not converge. Compare the harmonic series to a series where the first 4 terms are $1/4$, the next 8 terms are $1/8$, the next 16 terms are $1/16$, and so on.

Harmonic Series : $S = 1 + 1/2 + 1/3 + 1/4 + \dots$

Another Series : $T = 1/4 + 1/4 + 1/4 + 1/4 + (1/8 + \dots) + \dots$

The important thing to notice is the harmonic series has a sum at least as large as T . The reason is that the n^{th} term in the harmonic series is at least as big as the corresponding n^{th} term in the other series T (do a term by term comparison: 1 is bigger than $1/4$, $1/2$ is bigger than $1/4$, and so on). So whatever the sum of T is, the harmonic series should at least be as large.

What is the sum of the series T ?

We know that there are 4 terms of $1/4$, there are 8 terms of $1/8$, there are 16 terms of $1/16$, and so on. Therefore the sum of T is calculated as:

$$T = 4(1/4) + 8(1/8) + 16(1/16) + \dots = 1 + 1 + 1 + \dots = \infty$$

In other words, the series T has an infinite sum. And since the harmonic series has to sum to at least as large, that means the harmonic series also has an infinite sum. The sum grows very, very slowly, but remarkably the sum is unbounded.

Resolving the paradox

Here were the four steps that lead to the paradox.

$$(I.) S = 1 + 1/2 + 1/3 + 1/4 + 1/5 + \dots$$

$$(II.) \frac{1}{2} S = 1/2 + 1/4 + 1/6 + 1/8 + 1/10 + \dots$$

$$(III.) = (I.) - (II.) \frac{1}{2} S = 1 + 1/3 + 1/5 + 1/7 + 1/9 + \dots$$

$$(IV.) = (III.) - (II.) \frac{1}{2} S - \frac{1}{2} S = 0 = (1 - 1/2) + (1/3 - 1/4) + (1/5 - 1/6) + \dots$$

Where is the mistake? Note that since the harmonic series has an infinite sum, that means half of the harmonic series also has an infinite sum. In other words, both the series in (I.) and (II.) have an infinite sum.

Therefore, one flaw in the proof is step (III.), which is the subtraction of two series that have an infinite sum. Since it is nonsensical to subtract one infinite quantify from another, the resulting steps are incorrect, and that's why step (IV.) is an absurd conclusion.

Notes

1. This paradox comes from the website

http://www.clockstone.com/enigma/algebra_paradoxon_3.htm

2. Sloane's A082912 : Sum of $a(n)$ terms of harmonic series is $> 10^n$, *The On-Line Encyclopedia of Integer Sequences* . OEIS Foundation. Available online. <http://oeis.org/A082912> (Found via Wikipedia entry [http://en.wikipedia.org/wiki/Harmonic_series_\(mathematics\)](http://en.wikipedia.org/wiki/Harmonic_series_(mathematics)))

PROBABILITY PARADOXES

Think you understand randomness and chance? Take a stab at the probability problems in this section, which are famous for being both counter-intuitive and difficult to solve.

Some of the paradoxes demonstrate why it is a bad idea to rely on intuition (*Birthday Paradox*) .

In other times, the trouble is trying to understand why the correct answer makes sense, as seemingly immaterial details are relevant to find the odds (*Bertrand's Box Paradox*).

A third set of problems is about the nature of expectations and random sampling (the *St. Petersburg Paradox*).

Finally there are some problems that question the notion of probability and expectation (*Random Chord Paradox*) .

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15) Birthday Paradox

Is it really surprising, or was it bound to happen? The birthday paradox illustrates why some surprising coincidences are not really all that surprising.

The setup is the following problem. How many people are needed so there is a 50-50 chance that two people will have the same birthday? (It is assumed that each person is equally likely to be born on the 365 days in a standard year).

When most people hear this problem, they conclude the group would have to be pretty large. The thought process might go as follows.

Linear thinking

On one extreme, the chance is very low in a small group. In a group of 2 people, for instance, there is just a 1 in 365, or less than a one percent chance, that the two people would share the same birthday.

On the other extreme, the odds become very favorable in a large group. In fact, at some point the group would be sufficiently large so that two people would necessarily share the same birthday. A little bit of logical thinking will demonstrate the crucial group size is 366 people. In any smaller group, it could be possible that each person had a distinct birthday. In a group of 366 people, at most 365 could have different birthdays. The extra person would necessarily have to share a birthday with someone.

Let's recap the range. If a group of 2 people corresponds to about 0 percent chance of a shared birthday, and a group of 366 people translates to a 100 percent chance, it stands to reason that a group size somewhere in the middle would correspond to a 50 percent chance. That is, one might guess a group of about 180 people would “cover” half of the days in the year, achieving the 50 percent chance of a shared birthday.

Now comes the surprise. The ballpark answer is not just wrong, but it is not even in the same ballpark. The birthday problem cannot be solved by considering the linear trend .

The answer

The solution is a mere 23 people are required to achieve a 50 percent chance of two people sharing a birthday. The apparent paradox is trying to reconcile why the linear intuition does not work, and why the actual answer is so low.

Why is the answer 23? We have to take a step back and think about what the question is asking. When considering birthdays, people naturally focus on their own birthday. Based on personal experience, you would not expect someone in a group of 30 or 100 people to share your birthday. That would be a surprise.

But that's not what the birthday problem is asking. The question is not asking whether *you* would share a birthday with someone. The question is whether *any two people* might share the same birthday.

The answer becomes intuitively easier to understand when considering the pairs of birthdays in a group. With two people, there is just one pair of birthdays to compare. But with 3 people, there are 3 pairs of birthdays: persons 1 and 2, 2 and 3, and 1 and 3. With 4 people, there are 6 pairs of birthdays that might coincide (as in (1, 2) (1, 3) (1, 4) (2,3) (2,4) (3,4)). You get the idea.

Each time the group size increases, that extra person could potentially match with each of the previous group members. By the time the group size is 23, there are a possible 253 pairs of people who could share a birthday. At this point, there is a 50 percent chance that two people would share the same birthday.

Here is the formal calculation. The first person's birthday can be any of the 365 days. So that the second person does not match, it would have to be one of the remaining 364 days. Therefore, the chance the two do not share a birthday is $364/365$.

The third person's birthday cannot match *either* birthday of the previous two people. Hence, this person can only have a birthday on one of the remaining 363 days.

This logic can be extended for larger group sizes. More generally, the k^{th} person in a group could have a birthday on one of the remaining $(365 - k + 1)$ calendar days so as not to match with anyone else.

The joint probability that no one shares a birthday is the product of the individual odds. Hence, in a group of k people, the probability that *no one* in the group shares a birthday:

$$(364/365)(363/365)(362/365) \dots (365 - k + 1)/365$$

Since each of the terms has the same denominator of 365, and there are $k - 1$ terms, this expression can be rewritten as:

$$(364)(363)(362) \dots (365 - k + 1)/365^{k-1}$$

The chance that *someone* has the same birthday is the complement event, which is found by taking 1 minus the previous equation:

$$1 - (364)(363)(362) \dots (365 - k + 1)/365^{k-1}$$

The chance that two people share a birthday is still small for small groups (low values of k). In a group of 5 people, there is a 2.7 percent chance that two people will share a birthday. In a group of 10, this rises to a noticeable level of 11.7 percent. For each of the next 10 group sizes, each person added to the group increases the probability by about 3 percentage points. As the group gets larger, it becomes harder to avoid sharing a birthday; or in other words, it becomes more likely that two people will share a birthday.

In a group of 23, there is just over a 50 percent chance that two people will share a birthday.

Now, part of the paradox is that, in a group of 23, *you* in particular will not have a 50 percent chance of someone sharing your birthday. It is just some two people in the group will share a birthday.

The late-night TV host Johnny Carson famously misinterpreted the birthday problem, as detailed in John Allen Paulos' book *Innumeracy*. Carson noted there were over 100 people in the audience—much more than 23—so he

concluded someone in the audience must have his birthday. Evidently no one did, which made the viewer question the birthday problem answer .

But Carson was wrong in what he asked. If he had surveyed the audience to find if *any two people* shared a birthday, the result would have been almost surely yes. Instead, he was focusing on his particular birthday, which is an entirely different question.

How big a group is required so there is a 50-50 chance of sharing a specific birthday? This problem is a bit easier to solve and matches the intuition that a large group is needed.

The chance someone has *your* birthday

There are 364 birthdays that are not yours, so in a group of k people, the probability that none of the people shares your birthday is $(364/365)^k$. In other words, the chance that someone shares your birthday is $1 - (364/365)^k$. This equation can be set equal to 50 percent and solved to find that k should equal 253. In other words, you need at least 253 people so that someone matches your specific birthday with a 50 percent chance.

So while it might be relatively unusual that someone has your exact birthday, it is relatively unsurprising in a large group that two people would share a birthday.

16)False Positive Paradox

Suppose a rare disease afflicts 1 percent of the population, and the test for it is 99 percent accurate.

Fearing an outbreak, public health officials mandate tests for everyone in a town of 100,000.

The question is: if a person tests positive, what are the chances the person actually has the disease?

Because the test is 99 percent accurate, it might seem the results should be accurate too. In fact, the results are expected to be quite inaccurate: someone who tests positive will in fact have a 50 percent chance of *not* having the disease.

The question is, why does a very accurate test produce such inaccurate results?

Resolving the apparent paradox

The issue is the disease is rare. So even though the test will accurately identify people who *have* the disease, it will inadvertently *misdiagnose* healthy people as false positives. Here is the supporting calculation.

Of the 100,000 people in the town, the disease affects 1 percent, or 1,000 people. The test, which is 99 percent accurate, will identify 990 of the group as positive.

What about the healthy people? Of the 99,000 healthy people, the test will be 99 percent accurate. That means the test will identify 98,010 of the group as healthy. That leaves 1 percent, or 990 cases, that will be wrongly identified as false positives.

To summarize, the test will yield 1,980 positives: of which, 990 are correctly identified, and 990 are false positives.

So what does that mean? A large portion of people who tested positive—50 percent (990/1,980)—were false positives.

	<u>Total</u>	<u>Positive</u>	<u>Negative</u>
Healthy	99,000	990	98,010
Disease	1,000	990	10

While the test is 99 percent accurate, someone who tests positive only has a 50/50 chance of having the disease. The result is a consequence of the disease being rare. While the test can correctly identify people who have the disease, it will also misidentify many healthy people as false positives.

What the paradox teaches us

The problem illustrates why it might be necessary to run a test, even an accurate one, a second time. Using the same numbers above, if the test were run a second time, the 50 percent false positive rate would drop to just 1 percent. (Of the 990 false positives, only 10 would test as false positives a second time. In contrast, 980 of the 990 correct positives would test as positive a second time. So only 10 of the 990 people who tested positive two times would be false positives. This is a 1 percent rate of false positives. This level of accuracy is more in line with the stated 99 percent accuracy rate.)

Second, the problem suggests why mandatory screening could be a bad idea. For instance, the federal government could require that every couple get tested for various diseases before marriage. But given the millions of marriage every year, this blanket policy would probably result in an overwhelming amount of false positives, scaring many people for no reason.

What's the alternative to testing everyone? Doctors are cautious not to run a barrage of tests on everyone. They take medical histories and look for symptoms. They often administer tests to confirm their suspicion, not to screen for a problem. This is one reason why medical tests are often useful:

it's the doctor who does some initial screening to decrease the rate of false positives.

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17)Simpson's Paradox

This paradox is often explained with hospital survival rates. I discovered another example in America's college football rankings called the BCS ranking system.

Without getting into too many details, the BCS ranking broadly works like this. Teams are ranked independently by 3 different voting systems: the Harris Poll, the USA Today Poll, and a computer algorithm.

The 3 polls are then averaged, with each poll given equal weight. The final number, known as the BCS average, determines the overall ranking of teams.

The average is supposed to even out the biases of the various rankings. But several years ago I noticed something rather peculiar in the rankings.

Namely, I found this: a team that was ranked 6th or worse in all three polls ended up being ranked as the 5th overall team after the polls were averaged.

The weird thing I noticed occurred for the BCS rankings from October 26, 2008.

At the time, USC was ranked 7th according to the Harris Poll, 6th according to the USA Today Poll, and 6th according to the Computer Rankings.

And yet when all three rankings were averaged, USC ended up being ranked 5th overall. Here is the detailed table with the rankings¹.

		<u>Harris Poll</u>		<u>USA Today</u>		<u>Computer Rankings</u>		<u>BCS</u>
RK	TEAM	RK	%	RK	%	RK	%	AVG
1	Texas	1	.9961	1	.9980	1	1.000	.9981
2	Alabama	2	.9460	2	.9436	2	.960	.9499
3	Penn State	3	.9298	3	.9272	3	.920	.9257
4	Oklahoma	4	.8316	4	.8295	T-4	.820	.8270
5	USC	7	.7912	6	.8052	6	.750	.7822
6	Georgia	8	.7453	8	.7456	T-4	.820	.7703
7	Texas Tech	5	.8021	5	.8170	T-10	.610	.7431
8	Florida	6	.7923	7	.7862	T-10	.610	.7295

Here is a brief explanation of some of the numbers in the table:

–USC was ranked as 7th in the Harris Poll. Its Harris Poll average was 0.7912, behind Florida 6th (0.7923) and Texas Tech 5th (0.8021)

–USC was ranked as 6th in the USA Today Poll with an average of 0.8052.

–USC was ranked as 6th in the Computer Rankings with an average of 0.750.

–The three polls were averaged together to determine a BCS average. USC was the 5th highest number at 0.7822. The BCS average was calculated as the average of the three rankings, that is, $0.7822 = (0.7912 + 0.8052 + 0.750) / 3$

What is going on?

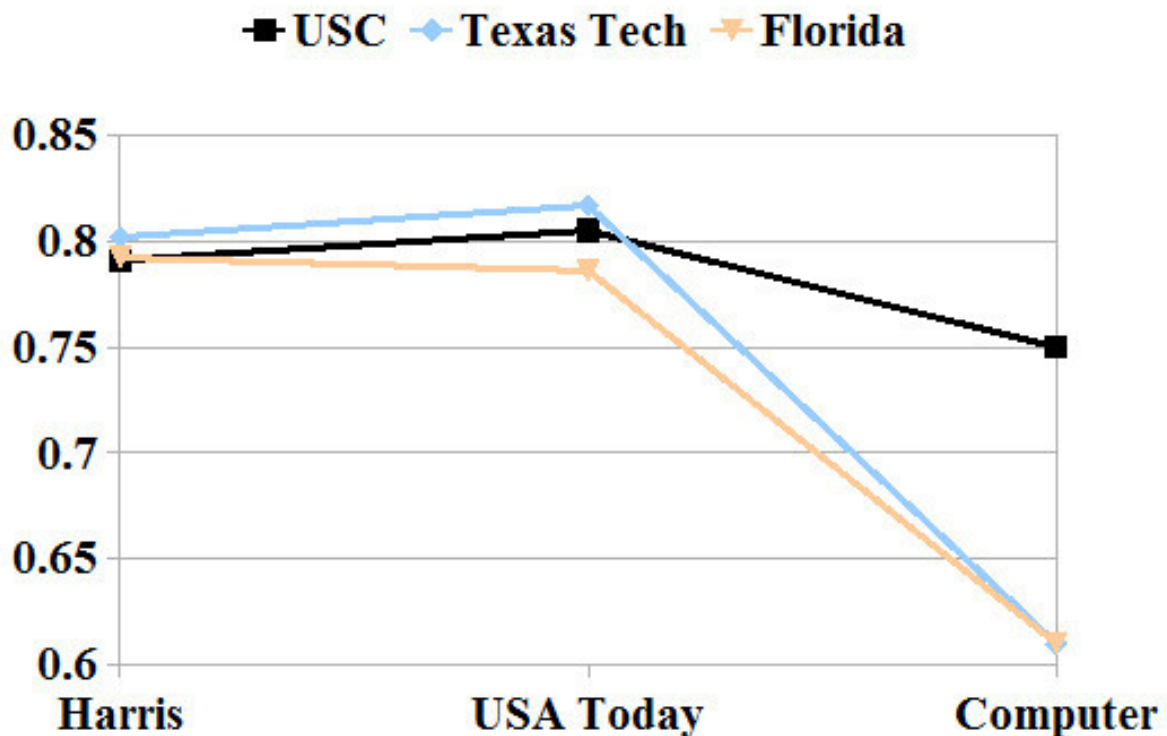
If USC was ranked no better than 6th in any individual poll, how could it have jumped up to 5th overall when the three polls were averaged?

The resolution

Basically USC was behind Texas Tech (and close to Florida) in the Harris and USA Today polls .

USC got a boost because its computer ranking—though it was 6th—was a large value. This basically let it leapfrog Texas Tech and Florida who got punished by the computers.

Here's an illustration of how USC had virtually the same ranking in other polls, but a large boost from the computer rankings.



The paradox is in thinking that each poll actually affects the overall ranking equally. This is not true: it is the total points that affect the overall average. Because the computer rankings punished Texas Tech, this skewed their overall average and dropped them behind USC.

Other examples

Simpson's paradox happens when combining averages for groups of different sizes. A canonical example is that one hospital can have better survival rates for cancer and heart patients, but it might be worse when looking at overall survival rates for both cancer and heart patients. The reason is the same as the football example: combining groups of different sizes can alter the overall average.

Similarly, it is theoretically possible for a baseball batter to have the highest average in both first and second halves of a season, but lose out to someone else in the overall batting average for a season.

Notes

1. The BCS rankings data were originally from the website ESPN.com. A cached copy of that data, with USC's data highlighted, is available here: <http://mindyourdecisions.com/blog/wp-content/uploads/2012/01/bcs-statistical-paradox.png>

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18) Bertrand's Box Paradox

Imagine I have three drawers. One drawer is filled with 2 gold coins, another is filled with 2 silver coins, and a third is filled with one of each coin.

Now I randomly pick a drawer from which I randomly select a coin. It turns out to be a silver coin. The question is, what's the chance the other coin is also a silver coin?

The paradox: it's not 50 percent!

Common sense is that the other coin could either be a silver coin or a gold coin, hence the answer should be 50 percent. But if that were the case, this problem really wouldn't be very interesting, and this wouldn't be a paradox.

The answer is the other coin has an overwhelming $\frac{2}{3}$ chance of being a silver coin. Huh? How come looking at a coin makes the other coin more likely to be the same type? We have to carefully consider the information we are given to see why that is the case.

Resolving the paradox

The trick is to carefully account for the possibilities. Let's label the coins as G for gold, g for a second gold coin, S for silver, and s for a second silver coin.

The logic is as follows: I begin by picking a drawer at random from the possibilities Gg , GS , and Ss . When I select a coin at random and it's silver, you can eliminate the possibility that I drew from the Gg drawer.

There are three equally likely events to consider.

- (1) I am showing S from the Ss drawer, and the other coin is s .
- (2) I am showing s from the Ss drawer, and the other coin is S .
- (3) I am showing S from the SG drawer, and the other coin is G .

In two of the three cases, the other coin is silver. Hence, the probability the other coin is silver is $2/3$.

The trick is that when you see a silver coin, it could really be one of two of the indistinguishable silver coins in the Ss drawer.

If you understand this problem, then the next paradox should be a cinch.

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19) Boy or Girl Paradox

You're doing a statistical analysis for the government. One of your co-workers is looking at a set of 10,000 parents who happen to have two children, with at least one boy.

Your co-worker poses a problem to you, "I picked a random family from the data set. What is the probability that both children are boys?"

(Another way this story is told. A stranger tells you she is the mother of 2 children. At least one of the children is a boy. What's the chance that both are boys?)

The paradox: it's not 50 percent!

This is one of the stranger paradoxes. We know the genetic odds of having a boy or girl are roughly 50/50. If someone tells us that one of their children is a boy, why should that affect our belief about the gender of the other child?

The answer is stunning because the information revealed changes the odds. It turns out there is just a $1/3$ chance that both are boys.

Resolving the paradox

A parent of 2 children could either have four possible scenarios of younger and older siblings, with B for boy and G for girl: BB, BG, GB, GG.

When you learn that at least one of the children is a boy, you can remove the GG possibility. Hence, there are three equally likely events to consider:

- (1) the parent has two boys
- (2) the parent has a boy with an older sister
- (3) the parent has a boy with a younger sister

In two of the three cases, the other sibling is a girl. Hence, the probability the child is a boy is $1/3$.

I admit I found this solution hard to believe at first. So I ran a test in my spreadsheet using a random number generator to check. I created a fictional data set of 10,000 families who had two children and at least one boy (I did this by creating about 14,000 families with two children and removing the ones with two girls.). I then counted how many of the families had two boys. The answer was 3,346, which means the chance that both children are boys is 33.46%, which is very close to the theoretical answer of $1/3$.

What this paradox teaches us

This is a somewhat controversial paradox as it depends on the exact phrasing. If I wrote, “the older child is a boy,” then we'd consider two possibilities:

- (1) the parent has a boy with a younger sister
- (2) the parent has a boy with a younger brother

These two events are equally likely, and so the answer would be $1/2$. In this paradox, it's the phrasing that is meant to be tricky, not the math.

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20)The Monty Hall Problem

This is perhaps one of the most famous probability puzzles. The problem gained notoriety when a reader Craig Whitaker posed it as a question to the genius Marilyn vos Savant in *Parade* magazine¹ .

The question is the following. Suppose you're on a game show called *Let's Make a Deal!* There are three doors; behind one is a car, and behind the other two are goats. You pick door 1, and the host opens door 3 to reveal a goat. The host asks if you would like to switch to door 2. Is it better to switch or stick with your initial choice?

Incidentally, the problem is named after the host of the show, Monty Hall.

The paradox: the odds are not even

We need to specify a few circumstances before we can solve this problem. We will assume that the prizes are placed behind each door at random, so that there is a $1/3$ chance that the car is behind each of the doors. We also need to make an assumption about the host's behavior. The host is not going to open a door and show you there is a car behind it. That would kill the suspense of the moment since it would mean you are either staying on a door that has a goat or switching to a door that has a goat. That's not very good TV, is it?

Even with this setup, it seems like it shouldn't matter if you switch or stay. After the host opens one of the doors and shows a goat, that means one of the doors has a car and the other has a goat. It would appear that there is a 50/50 chance that either door could have the car, and hence it doesn't matter if you switch or stay.

This reasoning sounds right but it is wrong. Remarkably you have a $2/3$ chance of winning if you switch. The reasoning is along the lines of *Bertrand's Box Paradox* . The idea is that the host opening the door does reveal some information.

One line of logic can demonstrate the soundness of switching. Note that if you stay, you only win if the initial door you picked had the grand prize.

This is a $\frac{1}{3}$ chance. On the other hand, if you switch, you would win if the grand prize were in *either of the doors you did not pick* . That's a $\frac{2}{3}$ chance.

Another way of seeing this is to write out the three possible scenarios and seeing which strategy wins. If:

–Grand prize is in door 1 ($\frac{1}{3}$ of time): you win if you stay on door 1.

–Grand prize is in door 2 ($\frac{1}{3}$ of time): the host will open door 3. You win if you switch but you lose if you stay on door 1.

–Grand prize is in door 3 ($\frac{1}{3}$ of time): the host will open door 2. You win if you switch but you lose if you stay on door 1.

We can add up these contingencies to conclude that you will win $\frac{2}{3}$ of the time if you switch, but you only win $\frac{1}{3}$ of the time if you stay.

Consider another thought experiment. You pick door 1 from doors 1 to 100. The host is ready to open 98 of the doors that you did not pick to reveal a goat behind each door. If you stay on door 1, you only win in the 1 in 100 chance that door 1 had a goat. If you switch, then you would win if the goat were on any of the 99 doors you did not pick.

The situation is paradoxical because it initially seems that the host opening the door doesn't change anything. The reality is that the host never opens the door with the grand prize. Hence you learn information when he opens the door, and you gain an edge by switching.

The problem is also a bit of a stain on the mathematical community. After Marilyn vos Savant wrote up the solution, many readers did not believe it was a $\frac{2}{3}$ chance. People wrote thousands of letters, including some people who had a Ph.D. in mathematics, that said she was wrong and the answer was $\frac{1}{2}$. And partly for this reason, the Monty Hall problem joined the ranks of famous math problems.

Notes

1. The problem appeared in the magazine *Parade* in 1990. Marilyn vos Savant has the problem and several critical letters listed her webpage.
<http://marilynvossavant.com/game-show-problem/>

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21)The Necktie Paradox

Two men are given neckties by their wives as Christmas presents¹. Over drinks they start arguing about who has the cheaper necktie.

They decide to make a playful wager to settle the matter. They will talk to their wives to find out which necktie is more expensive. The man with the more expensive necktie will be the loser, and additionally, he has to hand over his necktie as a prize.

How fair is this wager?

Is the other necktie always better?

The first man is thinking about the bet, and he suddenly feels that he has made a good wager.

He reasons as follows: “the probability of me winning or losing is an even 50/50 chance. If I lose, then I lose the value of my necktie. If I win, however, then I win *more* than the value of my necktie. That is, I can bet something worth x dollars and have a 50 percent chance of winning something valued more than x dollars. It's a winning proposition, and I should definitely make this wager!”

The second man, also thinking about the bet, uses the similar reasoning to arrive at the same conclusion. He feels the bet is a winning proposition, and he is happy to take the wager.

It is obviously not possible that both men could be advantaged in a zero-sum game. What is the reason for the necktie paradox?

Resolution 1: correct the logic

One of the problems is the men are reasoning incorrectly, a common downfall when trying to do math over drinks.

The issue is that each is considering the wager in terms of his own tie being the more expensive and less expensive tie *at the same time*. Clearly, one's

own tie has to be one or the other, either the more expensive or the less expensive .

Therefore, the reasoning should be:

–If I have the more expensive tie, and I make the bet, I will lose my more expensive tie, or

–If I have the less expensive tie, and I make the bet, I will win the more expensive tie

There is a 50/50 chance of being in either situation. Hence, there is no advantage to either player, and there is no mystery to the bet.

The game will end with each man winning half of the time, and hence the bet is fair.

Resolution 2: correct the expectation calculation

There is another way to view the problem. Suppose the more expensive necktie is worth x dollars. What is the expectation of the bet?

To calculate the expectation, you need to know two things.

First, what is the chance you win? This is the chance that your tie is more expensive. It is assumed that neither you nor the other man knows the value. Under such ignorance, it is reasonable to say there is a 50/50 chance that either tie is more expensive. Hence you win with a 50 percent chance and lose with a 50 percent chance.

Second, what is the payoff to the bet? In the case you win, you win the more expensive tie worth x dollars. In the case you lose, you have to give up your more expensive tie worth x dollars.

Putting these two facts together, we get:

$$E(\text{bet}) = \text{Pr}(\text{win})\text{gain} + \text{Pr}(\text{lose})\text{loss} = 0.5(x) + 0.5(-x) = 0$$

The math demonstrates the wager has an expected value of zero, that is, the bet is exactly fair to both players.

Resolution 3: never bet a gift from your wife

Of course, it should be clear the game is not really a zero-sum game but a negative sum game .

Upon learning the bet, and that their husbands would wager their thoughtful gifts, both wives will be angry. Clearly, there will be no winners, and the only safe bet is to avoid this game entirely. Consider yourself forewarned!

Notes

1. This description of the problem comes from the Wikipedia entry.
http://en.wikipedia.org/wiki/Necktie_paradox

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22)The Friendship Paradox

It's a mathematical fact. Your friends are probably more popular than you are.

Don't believe it? Consider this: the average Facebook user has 245 friends, but the average friend on Facebook has 359 friends, according to a 2011 Pew survey¹.

That's right. The average person on Facebook has fewer friends than their friends do.

But how can that be? It definitely seems weird, and that's why this phenomenon is known as "friendship paradox," described in a 1991 paper by Scott L. Feld amusingly titled "Why Your Friends Have More Friends Than You Do."²

The paradox is not so mysterious, however, after examining its emergence from mathematical facts.

The intuitive explanation

One way to think about this is to remember when you first joined Facebook. When you first created a profile, you started out with 0 friends. The quickest way to get friends was to add people who were already on Facebook. And since these people were already using Facebook, they already had a huge lead in friend count. And each time you added a friend, those people got to increase their friend count too! So clearly, it's not that hard to see why your friends would have more friends than you when you first joined.

Of course, as you invite more friends and make contacts you can certainly overtake people in the friend count. The friendship paradox is not about the time you joined.

It's about this: on average, you will tend to add friends of people who are popular because it's a social game.

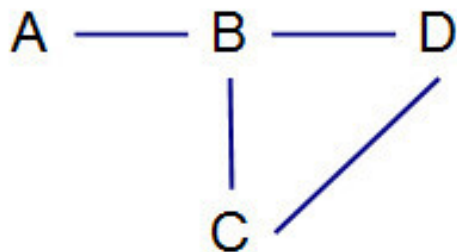
And in the end, the average Facebook user ends up having fewer friends than their friends do .

A concrete example

Let's consider a group of 4 people A, B, C, and D.

Imagine that A and B are friends, B is friends with everyone, C is friends with B and D, and D is friends with B and C.

We can draw a graph that illustrates the various friendships, where a connecting line means two people are friends:



Let's go through the calculation of counting friends, and counting friends of friends, to see if the friendship paradox holds.

A has just 1 friend, B has 3 friends, C and D each has 2 friends.

Now we will count the friends of friends. A is friends with B who has 3 friends. B is friends with everyone, which makes for 5 friends of friends. Similarly we will find C and D each has 5 friends of friends.

Here's a table that summarizes these statistics:

<u>Person</u>	<u>Friends</u>	<u>Friends of Friends</u>	<u>Average friends of friends</u>
A	1	3	3
B	3	5	1.67
C	2	5	2.50
D	2	5	2.50
Total	8	18	
Mean	2		2.25

You can see the average friends of friends column is higher for everyone but person B. So B is the lone exception: a very popular person will have more friends than their friends do (not a shock at all).

But now comes the interesting part when we look at the average. When we add up the total number of friends, we find there are 8 total friends amongst the 4 people. This means the average person has 2 friends.

On the other hand, we can find the total in the friends of friends column. Here, we will find there are 18 total “friends of friends” to be divided by the total of 8 “friends” relationships. This means the average friend has 2.25 friends.

In other words, the average person has 2 friends but the average friend has 2.25 friends. This is the friendship paradox.

Now let's prove why this happens.

The mathematical proof

We need to calculate two quantities: the mean number of friends for the entire graph, and the mean number of friends for a random person/node on the graph.

A little bit of notation can help. Let us say that person i has a number of friends x_i , and suppose there are n people.

What is the average number of friends in the entire graph? This is easy to calculate. We add up the total number of friends and divide by the total

number of people. The formula will be:

$$\mu = \frac{\sum x_i}{n}$$

We don't need to simplify that formula any more for our purposes. Now, we need to calculate the average friends of friends. This will be a bit harder, but not too bad.

The way we will do this is as follows. For each person, we will look at their friends and then add up the number of friends they have. So if B is friends with A, C, and D, we will need to add the numbers x_A , x_C and x_D .

The formula is a bit messy if you work person by person. Luckily, there is a trick.

We really only care about the total sum since this is an average. So we can ask the following question. For a given person i , how many times will the term x_i appear in our final summation?

The answer is easy. The only time we need to count friends of person i is for the friends of person i (I know that sounds confusing, read it again and it will make sense).

That means each of the friends of person i (there are x_i such friends) will contribute the term x_i to the final sum.

Hence, the final summation has the term $(x_i)(x_i) = x_i^2$.

The numerator will therefore be the sum of the squares of each person's friends. We need to divide this by the total number of friends which is the sum of x_i .

Therefore, the mean number of friends OF friends is:

$$\text{mean friend of friends} = \frac{\sum (x_i)^2}{\sum x_i}$$

Again, we do not need to simplify this term for our purposes.

What we will do is tinker with this formula so that we can compare this average to our previous one. The motivation is to rewrite the numerator with the squared sum in terms of other variables.

Once more, a statistical trick is involved. One has to remember the formula for variance, denoted σ^2 , of a discrete random variable is the following, where μ is the population mean:

$$\text{variance } \sigma^2 = \frac{\sum (x_i)^2}{n} - \mu^2$$

We can manipulate this equation to solve for the sum of the squared terms:

$$\sum (x_i)^2 = (\mu^2 + \sigma^2)n$$

And now, we will substitute for the sum of the squared terms in the expectation we wish to compute:

$$\frac{\sum (x_i)^2}{\sum x_i} = \frac{(\mu^2 + \sigma^2)n}{\mu n} = \mu + \frac{\sigma^2}{\mu}$$

We have found the average friends of friends is equal to the average friends PLUS another term. The other term is the variance of the number of friends divided by the average number of friends—both of these are non-negative in a typical social graph.

average friends $\leq$ average friends of friends

$$\mu \leq \mu + \frac{\sigma^2}{\mu}$$

So in the end, we have proven the average number of friends of friends will be equal or exceed the average number of friends.

An application: predicting who gets the flu

Researchers have found a way to apply the friendship paradox to the flu .

From a public health standpoint, one wants to vaccinate everyone to lessen the spread of the flu. But we know that is typically not possible. So another approach is to target people who are most likely to spread it. This will be someone who is social and has lots of friends.

Ideally, we could draw a graph of all the friendships and see who is a prime candidate. But in practice, that will be expensive and take a lot of time and money.

The researchers came up with this idea³ . First, take a random sample of people (like taking random nodes in a graph). Then, ask those people to name a friend.

Chances are, those people will not name an outcast. They will name someone that is popular. And by the above analysis, that person will likely be more connected and more likely to get the flu.

The study found that friends of the random sample did in fact get the flu earlier. This indicates friends of friends could be prime targets for getting vaccinated.

Notes

1. Number of Facebook friends: [Pew Survey](#) via The [Washington Post](#)

2. Why Your Friends Have More Friends Than You Do. Scott L. Feld.
American Journal of Sociology. Vol. 96, No. 6 (May, 1991), pp. 1464-1477.
Published by: The University of Chicago Press. Article Stable URL:
<http://www.jstor.org/stable/2781907>

3. [Infectious personalities](#) , The Economist.

[OceanofPDF.com](#)

23)St Petersburg Paradox

This chapter also appears in *Math Puzzles Volume 1* .

You are offered an unusual gamble.

A fair coin is tossed until the first heads appears, which ends the game. The payoff to you depends on the number of tosses. The payoff starts at \$2 and doubles on each successive toss.

That means you get \$2 if the first toss is a head, \$4 if the first toss is a tails and the second is a heads, \$8 if the first two tosses are tails and the third is a head, and so on. In other words, you get paid 2^k , where k is the number of tosses until the first heads.

<u>Toss # of first heads</u>	<u>Probability</u>	<u>Payout</u>
1	1/2	\$2
2	1/4	\$4
3	1/8	\$8
4	1/16	\$16
5	1/32	\$32
k	$(1/2)^k$	2^k

The question is how much should you be willing to pay to play this game? In other words, what is a fair price for this game?

The paradoxical part

The typical question is to compute the expectation. This is done by multiplying the various payouts by their probability of occurrence and adding it all together. To say it another way, the expectation is the weighted sum of the payouts given their likelihood.

The respective probabilities are easy to compute. The chance the first toss is a heads is $1/2$, the chance the first toss is a tails and the second is a heads is $(1/2)(1/2)$, and the third toss being the first heads is $(1/2)(1/2)(1/2)$, so the pattern is clear that the game ending on the k toss is $(1/2)^k$

Now, with probability $1/2$ you win \$2, with probability $1/4$ you win \$4, with probability $1/8$ you win \$8 and thus the expectation is:

$$E = (1/2)(\$2) + (1/4)(\$4) + (1/8)(\$8) + \dots = \$1 + \$1 + \$1 + \dots = \$\infty$$

The surprising result is the expectation is infinite. This means this game—if played exactly as described—offers a potentially unlimited payout. In theory, a rational player should logically be willing to pay an astronomical amount to play this game, like paying a million dollars, a trillion dollars, and so on.

The fair price of infinity is paradoxical because the game at first does not seem very favorable. Few people would be willing to pay more than a couple of dollars to play this game, let alone \$100 or \$1,000.

But expectation theory seemingly says that any amount of money is justifiable. Banks should be willing to offer loans so people could play this game; venture capital firms should offer more money than they do to start-ups; individuals should be willing to mortgage their house, take a cash advance on their credit card, and take a payday loan.

What's going on here? Why is the expectation theory fair price so different from common sense? It turns out there are a variety of explanations.

Resolution 1: Payouts should be realistic

Imagine you are playing this game with a friend. You hit a lucky streak. The first nine tosses have been tails and you're still going. If the tenth toss is a heads, then you get \$1,204 as a payout. If it's a tails, you have a chance to win \$2,408, and even more.

At this point your friend realizes he's made a mistake. He thought he'd cash out with your \$10 entry fee, but he now is in trouble and he cannot afford to

pay any more.

He pleads with you to stop. He'll gladly pay you the \$512 you've earned—so long as you keep this whole bet a secret from his wife. What would you do in this situation?

Most of us would take the cash and show some mercy. There is no joy in winning if it means crippling a friend financially. And this concocted scenario leads to one of the unrealistic assumptions of the St. Petersburg Paradox.

In the hypothetical coin game, you're supposed to believe the other side can pay out infinitely large sums of money. It doesn't happen often, but if you get to 20 coin tosses, you fully expect to be paid \$1,048,576.

This is unrealistic if you're playing with a friend or even a really, really rich friend. It might be possible with a casino, but even a casino may have a limited bankroll.

The truth is that payouts cannot be infinite. If such a game were to exist, there must be a maximum, finite payout.

This means the expectation is not an infinite sum but rather a sum of a finite number of terms. Depending on the size of the bankroll, the gamble has a finite payout.

Here are a few calculated expected values when considering a bankroll with a maximum payout¹ :

Backer	Bankroll	Expected value
Friendly game	\$100	\$7.56
Rich	\$1,000,000	\$20.91
Very Rich	\$1,000,000,000	\$30.86
Bill Gates (2013*)	\$67,000,000,000	\$36.95

*source: *Forbes*

Even in the unlikely event you are playing against Bill Gates, who could pay out an amazingly high amount, the gamble could not possibly be worth more than \$37.

As you can see, with a more realistic model of the game, the gut feeling that the game is worth a few dollars matches up with the math of the gamble.

Resolution 2: diminishing marginal utility

A quantity like \$1,000 has meaning to most people. If you were to ask a friend for such a loan, they would ask how you can pay it back, what you would use it for, and so on.

But there are times when \$1,000 seems to lose its value. I like to think about the show [*Deal or No Deal*](#) where contestants play a multi-stage lottery to win \$1,000,000. At various points in the game the contestants can either keep pursuing the big prizes or they can accept smaller consolation prizes. As the prospect of a big prize increases, the contestants start to care less and less about smaller prizes like \$1,000.

This is an example of the famous concept of diminishing marginal utility—the idea that at larger levels of consumption, incremental units are worth less. The concept is applicable for wealth decisions because at some point incremental earnings mean less to a person.

What this means for the gamble is that the payouts should be altered. The payouts should not be measured in dollars but rather by the utility the money provides.

One way to model utility is to use a logarithm function. Instead of saying the payout for the first toss is \$2, we will say it is $\log(2)$ units of utility, and accordingly adjust a payout of $\$x$ to be a utility of $\log(x)$.

Using a log utility function, the St Petersburg game now has a finite payout. Here is a derivation:

$$E = (1/2) \log 2 + (1/4) \log 4 + (1/8) \log 8 + \dots$$

$$E = (1/2) \log 2 + (2/4) \log 2 + (3/8) \log 2 + \dots$$

Let $u = \log 2$. Then ,

$$E = (1/2) u + (2/4) u + (3/8) u + \dots$$

$$- (1/2) E = - (1/4) u - (2/8) u + \dots$$

Therefore,

$$(1/2) E = (1/2) u + (1/4) u + \dots$$

$$1/2 E = u$$

$$E = 2u = 2 \log 2$$

This is a small payout but the actual quantity does not matter. The point is the payout is finite, which again resolves the original problem.

The St. Petersburg Paradox is interesting in that a simple game could yield a mathematical infinite payout. But in reality, the game would have a finite expected value. Thus, this matches the intuition that one should only pay about a fistful of dollars to play this game.

Notes

1. The table with maximum payouts for a given bankroll was adapted from a table on Wikipedia [http://en.wikipedia.org/wiki/St. Petersburg_paradox](http://en.wikipedia.org/wiki/St._Petersburg_paradox)

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24) Wild Card Poker Paradox

Wild card poker is a variation of poker that implements the use of at least one "wild card." A wild card is a designated card that a player can assign any value and suit. A wild card livens the action by making stronger hands easier to complete.

This section is about the following question. In wild card poker, which hand is more valuable: three of a kind or two pair?

It turns out the question is something of a paradox. Before I get into this question, it is useful to understand the logic of poker hand rankings.

What makes a poker hand valuable?

Poker hand rankings could be arbitrarily chosen and simply based on tradition.

But, as everyone knows, there is a clever logic to poker rankings. The value of a hand is determined by its mathematical probability of occurrence. A hand is ranked *more* valuable if it is *less* likely or harder to happen by chance.

This is why a flush has a higher ranking than a straight: it is harder and less probable to complete five unsequenced cards of a suit (flush) than to get five sequenced but unsuited cards (straight).

It is a standard probability exercise to verify that the poker hand rankings are based on their frequency of occurrence. Here is a table that summarizes the rankings and probabilities¹ :

RANKING OF POKER HAND VS PROBABILITY		
<u>Poker hand</u>	<u>Number of ways</u>	<u>Probability</u>
Straight Flush	40	0.0015%
Four of a Kind	624	0.024%
Full House	3,744	0.14%
Flush	5,108	0.20%
Straight	10,200	0.39%
Three of a Kind	54,912	2.10%
Two Pair	123,552	4.75%
One Pair	1,098,240	42.0%
High Card	1,302,540	50.1%
Total	2,598,960	100%

In a way, the frequency ranking makes the game of poker fair and logical. A three of a kind is more valuable than a two pair exactly because it is harder to achieve by about two percentage points.

This ranking is fundamental to standard games of poker. But an extension is to ask: what happens when a wild card is introduced?

Wild card poker effects

Poker can also be played with wild cards to liven up the action. One common variation is to add a joker card to the deck and allow it to be wild and represent any card.

The addition of a wild card does three things.

1. It creates a new hand "five of a kind" where all five cards are the same value, like five aces or five kings.
2. It affects the probability of getting a hand. It is now easier to make stronger hands because the wild card can be designated to be any card. A player with a wild card can always, for instance, make a pair .
3. It forces a player to make a choice about the wild card. Which card should the wild card represent? The answer is obviously to make the choice that gives the best possible hand with the other cards.

This third effect creates a new choice for the player in declaring a hand. If a player holds a pair and also a wild card, then there are two possible ways to declare the hand. The hand can be called a three of a kind if the player declares the wild card to match the pair. Alternately the hand can be called a two pair if the player declares the wild card to match an unpaired card to go along with the pair.

(Say a player has 4,4,2,6, and a joker. The joker can either be a 4 to complete a three of a kind 444, or it can be a 6 to complete two pairs of 44 and 66)

Which hand should be chosen? This is the source of the paradox.

The wild card poker paradox

For the moment we will assume the standard ranking that three of a kind is more valuable than a two pair.

We can then calculate the probability of making each hand in wild card poker. We will make the assumption that a player holding a pair and a wild card will choose to make this hand a three of a kind rather than a two pair.

The resulting probabilities are² :

SINGLE WILD CARD POKER (THREE OF A KIND RANKED ABOVE TWO PAIR)		
<u>Poker hand</u>	<u>Number of ways</u>	<u>Probability</u>
Five of a Kind	13	0.00045%
Straight Flush	184	0.0064%
Four of a Kind	3,120	0.110%
Full House	6,552	0.23%
Flush	7,804	0.20%
Straight	20,532	0.27%
Three of a Kind	137,280	4.80%
Two Pair	123,552	4.75%
One Pair	1,268,088	44.0%
High Card	1,302,560	45.0%
Total	2,869,685	100%

Most of the frequencies are in line with the standard ranking of poker hands, but there is one glaring exception. The three of a kind occurs with a slightly *higher* frequency than two pair! This violates the principle that less probable hands are ranked as more valuable.

This table suggests that ranking three of a kind higher than two pair is a mistake. One attempt to fix this problem is to reverse the rankings: instead make two pair more valuable than a three of a kind.

The problem with this approach is that players will respond to the rule change. If a player is holding a pair and a wild card, then the player will

declare the wild card to be the higher ranked two pair. Therefore, we have to recalculate the probabilities.

Under this new rule, the probabilities become³

SINGLE WILD CARD POKER (TWO PAIR RANKED ABOVE THREE OF A KIND)		
<u>Poker hand</u>	<u>Number of ways</u>	<u>Probability</u>
Five of a Kind	13	0.00045%
Straight Flush	184	0.0064%
Four of a Kind	3,120	0.110%
Full House	6,552	0.23%
Flush	7,804	0.20%
Straight	20,532	0.27%
Two Pair	205,920	7.20%
Three of a Kind	54,912	1.90%
One Pair	1,268,088	44.0%
High Card	1,302,560	45.0%
Total	2,869,685	100%

Again, the table is fine except for one exception: now that the two pair is ranked as more valuable, the frequency it occurs is larger than that for three of a kind! The poker rankings again violate the principle that the valuable hands should occur less frequently.

As one can imagine, the ranking problem persists when playing wild card poker with two jokers or other rules like "deuces wild." The situation was

studied by mathematical researchers John Emert and Dale Umbach of Ball State University in a 1996 *Chance* article who explained⁴ : “When wild cards are allowed, there is no ranking of the hands that can be formed for which more valuable hands occur less frequently.”

The paradox is not going away, and it is fruitless to pursue a completely logical frequency ranking in wild card poker.

Possibly resolving the paradox: inclusion frequency

Emert and Umbach did take a stab at fixing the paradox. Since frequency ranking could not work, it would have to be abandoned for a new standard.

The idea is called inclusion frequency, and they prove this can eliminate the paradox. The problem is that inclusion frequency swaps the ranking of two hands from the traditional rankings. Under the inclusion frequency ranking, the flush becomes more valuable than a full house when the game has one or more wilds. With four wilds it can be shown that a flush is more valuable than even a four of a kind!

You can read more about the technical aspect of the inclusion ranking online⁵ .

While this new ranking assigns higher rankings to hands that occur less frequently, it comes at the price of changing the traditional rankings of the cards. This would invariably be confusing to players who have logged thousands of hours with the traditional ranking of hands. Imposing the new ranking would be on the annoyance level of swapping two keys on a keyboard—it would be painful and hard to learn the new rules.

And so, ranking cards in wild card poker remains something of a problem.

Notes

1. A detailed derivation of poker hands is available on Wikipedia http://en.wikipedia.org/wiki/List_of_poker_hands

2. The ranking of hands with a wild card is listed on the website
<http://www.curiouser.co.uk/paradoxes/wildcard.htm>
3. Same as 2.
4. Quoted in Peterson, Ivars *Mathtrek* . “Trouble with Wild-Card Poker.”
http://www.maa.org/mathland/mathland_9_9.html
5. The inclusion-frequency rankings are listed at the website
<http://www.curiouser.co.uk/paradoxes/wildcard2.htm>

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25) Gambler's Fallacy

You're at the casino playing poker and you've been getting bad cards all night. But you keep your hopes up high, thinking, "Things have to even out, I'm sure to get a good hand soon."

A star basketball player has been cold all night. He normally makes about 50 percent of his shot attempts, but tonight he is in a slump and he has missed all 10 of his shots. The announcer notes, "He's just got to keep shooting his way through it; he's bound to get out of his cold streak."

Your friend has flipped a coin and 5 times in a row it has shown heads. You feel like the next few flips should be disproportionately tails. It has to even out to a 50/50 average, right?

Alas, each of these intuitions is wrong. Each poker hand dealt is a new draw and is in fact more likely to be a poor hand than a good hand. Similarly, each shot attempt by a basketball player is more or less a statistically independent shot: there seems to be no hot or cold streaks¹. And coin flipping is also an independent event. Each flip is equally likely to be heads or tails.

The paradox is the conflict of two true statements. One statement is that we know things have to average out over time, a phenomenon known as the law of large numbers. If you play enough poker hands, take enough shot attempts, and flip a coin long enough, the outcomes will neatly fall into the average distributions.

The other statement is that each trial is independent. So if you get a string of coin flips that are tails, your next flip will equally be heads or tails.

So here's the paradox: if you start out with a string of tails, and things have to even out, why shouldn't the remaining tosses be favored to be heads? If that's not the case, how are things going to reach the average?

Resolving the paradox

The issue is one of perspective. Let's say you flip 50 tails by pure chance. Now you might think that the next 50 tosses should be favored to be heads. But that's only if you think the distribution needs to meet the average "soon." The truth is, the law of large numbers applies *in the long run*. It might be the case that the distribution might not average out for 200 flips, or maybe 1,000 flips, or even 10,000 flips.

In the short-run, we can have long streaks favoring one outcome or another. The fallacy is confusing the long-run convergence with our desire to have results even out soon.

Notes

1. Gilovich, Thomas, Robert Vallone, and Amos Tversky. "The hot hand in basketball: On the misperception of random sequences." *Cognitive Psychology* 17.3 (1985): 295-314.
<http://www.psych.cornell.edu/sec/pubPeople/tdg1/Gilo.Vallone.Tversky.pdf>

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26) Surprise Quiz Paradox

In high school, a teacher announced he would give a pop quiz during the next week. I just laughed. Tricks are for rabbits, silly teacher; you can't surprise a rational student.

Let's see why using systematic reasoning.

Could the quiz ever be a surprise on Friday, the last day of the week?

No, it cannot. If I happened to make it until Thursday night without taking a quiz, I would think, "Hey, the professor has not given the quiz yet. Since tomorrow is the last class of the week, the quiz has to occur on Friday. I won't be surprised when he pulls out the quiz."

So Friday is ruled out. I might then ask: could the quiz ever be a surprise on Thursday?

Again, it cannot. If I made it to Wednesday night without being quizzed, I would think, "The quiz can either be on Thursday or Friday. But Friday is ruled out. Therefore, the quiz has to be on Thursday. I won't be surprised."

And so Thursday is ruled out. Repeating this logic, I could then rule out Wednesday, Tuesday, and Monday as possible days when the quiz would be a surprise.

It is logically impossible for the teacher to give a surprise quiz.

And yet, if the teacher randomly gives the quiz during the week, I will not be able to predict it.

How is it possible that I cannot be surprised, and yet the teacher can surprise me?

What gives?

The surprise quiz paradox is a special case of the "unexpected hanging paradox," which has the same logical setup for a prisoner to be executed on

a “surprise” day. There are two explanations for the apparent paradox .

The first is that not everyone is a computing machine. It took five steps of reasoning to eliminate the days of the week. Most of us get bored after thinking one or two steps, which is why most of us can only anticipate a couple moves ahead in chess.

The second is a finer, epistemic point. Although teachers cannot surprise me with a quiz, that doesn't mean I can predict when the quiz will be. This is because the phrases "not surprised" and "able to predict" are two different concepts. For instance, when I toss a fair coin, I am not surprised if it turns up heads. But that doesn't mean I am able to predict the outcome of any particular flip.

The technical way to say it is that an event you cannot predict is the *realizations of a random process* . A head or a tail is a realization of a coin flip. A true surprise is a realization that's "vastly different" from what you expected. Like if a coin landed directly on its side, ten times in a row.

Similarly, I am not at all surprised when the market swings up or down. Fluctuations are a normal part of the market, and it's very difficult if not impossible to consistently predict financial markets.

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27)Two Envelopes

I have two envelopes A and B. One of them contains twice as much money as the other. Which envelope do you want?

Let's say you select envelope A and you look inside. Now, I ask you this: would you like to switch to envelope B, or stick with your initial choice?

In your mind, you do the following calculation. Imagine that envelope A has x dollars. Since one of the envelopes has twice as much money, that means envelope B is either the smaller amount ($x/2$) or it is the larger amount ($2x$). You have no reason to believe it is one or the other, so each event is equally likely.

What's the expected value to switching? That would be the average of the two possible values:

$$E(\text{switch to B}) = 0.5 (x/2) + 0.5 (2x) = 1.25x$$

So you decide the right choice is switching to envelope B. Now comes the paradoxical part.

There was really no reason you picked envelope A at the beginning. In fact, let's imagine that you picked envelope B. Should you stay or switch?

By exactly the same reasoning, you can evaluate the options. If envelope B has y dollars, then envelope A could either have less ($y/2$) or more ($2y$) with equal probability. Hence, you could reason the expected value of switching to A is:

$$E(\text{switch to A}) = 0.5 (y/2) + 0.5 (2y) = 1.25y$$

Evidently, if you started with envelope B, it is correct to switch to envelope A.

And that is the paradox: it seems that whichever envelope you picked first, it is better to switch to the other one! But that is impossible: how can it be logical to desire to think both envelope A and envelope B are favorable ?

Resolving the paradox

The main practical issue is in the expectation. You know that one envelope contains twice the amount of the other, but you probably have some idea of the range.

For instance, if you played this game with your parents, you might know that the most they could possibly give you is a tax-free gift of \$10,000. If you look inside and find the envelope is \$10,000, then there is no way you'd switch: you'd have a strong belief that you already got the bigger amount.

In mathematical jargon, the idea that you are completely ignorant of the amount in the envelope is an *improper prior*, or an *improper probability distribution* .

A practical solution is that you'd peek inside the envelope and have some idea of what the maximum amount is. If the amount is large enough (say more than half of the maximum), you are going to be pretty sure that you have selected the envelope with more money. Otherwise, the expectation above indicates that switching is probably favorable.

There is also the issue of risk. Let's say that you open an envelope and find that it has \$1 million. You are ecstatic about that amount. Do you risk that amount? Do you really want to take a 50/50 chance of dropping down to \$500,000 just for the chance of going up to \$2 million? Most of us probably would not, as it seems a bit greedy to be unhappy at \$1 million.

28) Siegel's Paradox of Exchange Rates

Exchange rates always seemed simple to me, until I learned a bit more finance. There are some quirky things that can happen **like this paradox**.

Imagine Bob frequently travels between the U.S. and Canada, so he keeps an eye out on the exchange rate.

Currently the exchange rate is 1 (\$/C\$), so Bob can trade 1 dollar U.S. for 1 dollar Canadian.

The market expects that in one year there is a 50 percent chance the U.S. dollar will be stronger by 10 percent, and a 50 percent chance the U.S. dollar will be weaker by 10 percent.

Question: what is the expected exchange rate one year from now? Solve the problem for both a U.S. and Canadian investor.

Siegel's paradox

If Bob lived in America, he would see the problem as follows. There is a 50 percent chance the exchange rate will be 1.1 (\$/C\$), and there is a 50 percent chance the exchange rate will be 0.9 (\$/C\$).

Thus, the expected exchange rate one year from now would be:

$$E(\$/C\$) = (0.5)(1.1) + (0.5)(0.9) = 1$$

As an American investor, Bob expects the future exchange rate to be 1 (\$/C\$).

But what if Bob lived in Canada? How would he see the problem?

In that case, he would have the following numbers in terms of Canadian dollars per U.S. dollar. There is a 50 percent chance the exchange rate will be $1/1.1$ (C\$/\\$), and there is a 50 percent chance the exchange rate will be $1/0.9$ (C\$/\\$).

Thus, the expected exchange rate one year from now, in \$/C\$, would be this somewhat cumbersome expression:

$$1 / E(C\$/\$) = 1 / [(0.5)(1/1.1) + (0.5)(1/0.9)] = 0.99$$

Note that if Bob lived in America, he expects the future rate to be 1 (\$/C\$). But if he lived in Canada, using the same assumptions, the future rate would be 0.99 (\$/C\$).

The exchange rates are very close, but they are not the same! Why is there any difference? This is the quirky result known as Siegel's paradox.

Why Siegel's paradox happens

There is a mathematical explanation for why the two expectations are not equal.

Specifically, if the exchange rate is the random variable X , then the American investor calculates $E(X)$. The Canadian investor is calculating $1/E(1/X)$.

These two values will not be equal due to something called Jensen's inequality¹, which implies that

$$E(X) > 1/E(1/X)$$

or alternately $E(1/X) > 1/E(X)$

The math explanation is the easier part of resolving the paradox. The bigger question is whether investors can profit on this information.

The long and short is there is no real economic impact of Siegel's paradox. It turns out to be a monetary illusion. What happens in real markets is that prices and interest rates in both countries typically adjust to prevent profitable anomalies².

Notes

1. In probability theory, Jensen's inequality states that for a convex function u , the following expectation holds: $u(E[x]) \leq E[u(x)]$

http://en.wikipedia.org/wiki/Jensen's_inequality.

2. For more on Siegel's paradox, consult the following paper. Aaron S. Edlin. "Forward Discount Bias, Nalebuff's Envelope Puzzle, and the Siegel Paradox in Foreign Exchange" *Topics in Theoretical Economics* 2 (2002). Available at: http://works.bepress.com/aaron_edlin/29

I first learned about Siegel's paradox in the first chapter of the book *Puzzles of Finance* by Mark Kritzman (Wiley 2000).

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29)Allais Paradox

This is a fun example in decision theory that demonstrates how our preferences can be inconsistent. As a point of trivia, Maurice Allais won the Nobel Prize for other research, but he is most remembered for this paradox.

The paradox is best illustrated with a brief experiment. The following questions are about choosing between a pair of gambles. Consider each choice and decide which you prefer.

choice a)	choice b)
\$1 million for sure	\$5 million with 10% chance \$1 million with 89% chance \$0 with 1% chance

Do you prefer choice a) or choice b)?

Now try experiment 2.

choice c)	choice d)
\$1 million with 11% chance \$0 with 89% chance	\$5 million with 10% chance \$0 with 90% chance

Do you prefer choice c) or choice d)?

While there are no right or wrong answers to the experiments, there should be some consistency between the two choices. For instance, if someone said they preferred \$100 over 2 movie tickets, then it would be reasonable to halve each amount and conclude the person prefers \$50 over a single movie ticket.

The Allais paradox is about finding the inconsistency between the answers between experiments 1 and 2 .

What do people pick?

In experimental results, many people choose a) for experiment 1 and d) for experiment 2. This seems to make reasonable sense. In the first experiment, taking the \$1 million for sure seems quite sensible. In the second, there is just a 1 percent extra chance of getting \$1 million, which most people will trade for prospect of gaining \$5 million.

Reasonable as these choices seem, they are not consistent with rational decision theory. That is, the answers are like someone choosing \$100 over 2 movie tickets but then preferring 1 movie ticket over \$50.

Where is the inconsistency?

The basis of decision theory is to translate amounts like \$1 million or \$5 million into a personal level of enjoyment, known as utility. The amounts x are converted by the utility function $u(x)$. The utility function itself is not important, but it should follow basic algebra rules. For instance, if $u(100) > u(50)$, then taking one half of each side should result in $0.5 u(100) > 0.5 u(50)$.

The following tables translate the experiment choices into expressions of utility.

Experiment 1

choice a)	choice b)
\$1 million for sure	\$5 million with 10% chance \$1 million with 89% chance \$0 with 1% chance
$u(\$1 \text{ M})$	$0.10 u(\$5 \text{ M}) + 0.89 u(\$1 \text{ M}) + 0.01 u(\$0)$

Experiment 2

choice c)	choice d)
\$1 million with 11% chance \$0 with 89% chance	\$5 million with 10% chance \$0 with 90% chance
$0.11 u(\$1 \text{ M}) + 0.89 u(\$0)$	$0.10 u(\$5 \text{ M}) + 0.90 u(\$0)$

In experiment 1, preferring the choice a) over choice b), logically implies that the utility from choice a) is a larger number than the utility from choice b). In other words, the mathematical expression is the following: $u(\$1 \text{ M}) > 0.10 u(\$5 \text{ M}) + 0.89u(\$1 \text{ M}) + 0.01 u(\$0)$

Now consider the results of experiment 2. If someone prefers d) over c), then that means the following mathematical expression: $0.11 u(\$1 \text{ M}) + 0.89 u(\$0) < 0.10 u(\$5 \text{ M}) + 0.90 u(\$0)$

So far, everything has been straightforward. The next step is to re-arrange the mathematical equations using algebra from experiment 2. The first step is to subtract the term $0.89 u(\$0)$, and then the second step is to rewrite the decimal 0.11 instead as the term $1 - 0.89$. Finally, the equation can be rearranged to reach the conclusion.

The algebra has been omitted, but here are the conclusions that can be reached.

Result from experiment 2 (after algebra)

$$u(\$1\text{ M}) < 0.89 u(\$1\text{ M}) + 0.10 u(\$5\text{ M}) + 0.01 u(\$0)$$

Result from experiment 1

$$u(\$1\text{ M}) > 0.89 u(\$1\text{ M}) + 0.10 u(\$5\text{ M}) + 0.01 u(\$0)$$

That was a lot of algebra, but now compare the results of the two experiments. These two are directly contradicting statements! And therein lies the paradox: in the first experiment the \$1 million for sure was preferred, but in the second the gamble for \$5 million was preferred.

At the heart of the paradox is the independence axiom of expected utility theory, which says that if a person prefers choice A over choice B, then the person should also prefer the same choices when they are mixed with some other event at the same probability p [person should prefer $p A + (1 - p) C$ to $p B + (1 - p) C$]

By this logic, the only choices consistent with decision theory are the pairs a) and c) or the pairs b) and d), and yet that is not what people generally prefer. Why not?

Explaining the paradox

There have been many attempts to explain why experimental results are inconsistent with theory.

One explanation that seems sensible is the experiments deal with extreme events of very low and very high probability. In these situations, people may not perceive risk perfectly according to rational decision theory, and there can be biases towards prospects framed in favorable ways.

For example, getting \$1 million for sure is a very favorable prospect, but when that probability is reduced from certainty to a lottery of 11 percent, as in experiment 2, it seems okay to take a gamble and pursue the greater gain of \$5 million in a lottery of 10 percent.

30) Ellsberg Paradox

Suppose I have two decks of cards:

	Deck A	Deck B
Red	24	?
Black	26	?
Total	50	50

Deck A contains 24 red cards and 26 black cards. Deck B contains 50 total cards, with red and black cards, but you don't know how many of each.

I randomly select a card from each deck. I offer you a chance to win some money; how would you answer each question?

Choice 1 (Black wins) : you get \$100 if the card you pick is *black* . Do you want the card from deck A or B?

Choice 2 (Red wins) : you get \$100 if the card you pick is *red* . Do you want the card from deck A or B?

Think about this before reading how people typically play.

The results

In choice 1, most people pick the card from deck A. This seems reasonable as deck A is known to have more black cards than red cards.

In choice 2, again most people pick the card from deck A. The choice feels intuitively right: having a certain 24/50 chance of winning seems like good odds, especially since the distribution of cards in deck B is unknown .

In many variations of this setup¹ , the results show that people choose from the known probability distribution in both situations. But this, it turns out, is not a mathematically correct choice. Do you see why?

The Ellsberg Paradox

In choice 1, you win if the card selected is black. In choice 2, you win if the card selected is red. What is the correct thing to do?

The trouble is you don't know how many red or black cards are in deck B. Nevertheless, you can reason that one of the following must be true about deck B:

- (I) It has equal red and black cards (25 red, 25 blue)
- (II) It has more red cards (at least 26 red)
- (III) It has more black cards (at least 26 black)

If you think (I) or (II) is true, then deck B is the better, or at least as good, for choice 2 when a red card wins. If you think (III) is true, then deck B is better, or at least as good, for choice 1 when you need a black card to win.

The correct thing to do when...		
	<u>Choice 1</u> <u>Black wins</u>	<u>Choice 2</u> <u>Red wins</u>
Deck B has	(I) Red = Black	Deck A
	(II) More red	Deck A
	(III) More black	Deck B

In other words, it always makes sense to pick deck B for one of the choices and deck A for the other.

(Technically both decks could have 26 black and 24 red, in which case it would still be correct to pick deck B in one of the choices).

The fact that deck A is picked in both situations is a decision bias known as the Ellsberg Paradox. It illustrates that we have a bias towards known distributions, even when uncertain distributions would logically be favorable .

This behavior seems sensible when the unknown risks could lead to new problems (if you switch internet providers, you might face unexpected blackouts or bad customer service).

But it also suggests a tendency to stick with a bad choice just to avoid trying something new, like:

- Considering new insurance options
- Trying a generic instead of a brand name
- Sticking with a bad investment versus looking for a new option

They say if it isn't broken then don't fix it. The Ellsberg Paradox demonstrates how we often don't even fix something that is broken.

Notes

1. [How Much Ambiguity Aversion?](#) by Binmore, Stewart, Voorhoeve and [Risk, Ambiguity, and the Rank-Dependence Axioms](#) by Mark J. Machina.

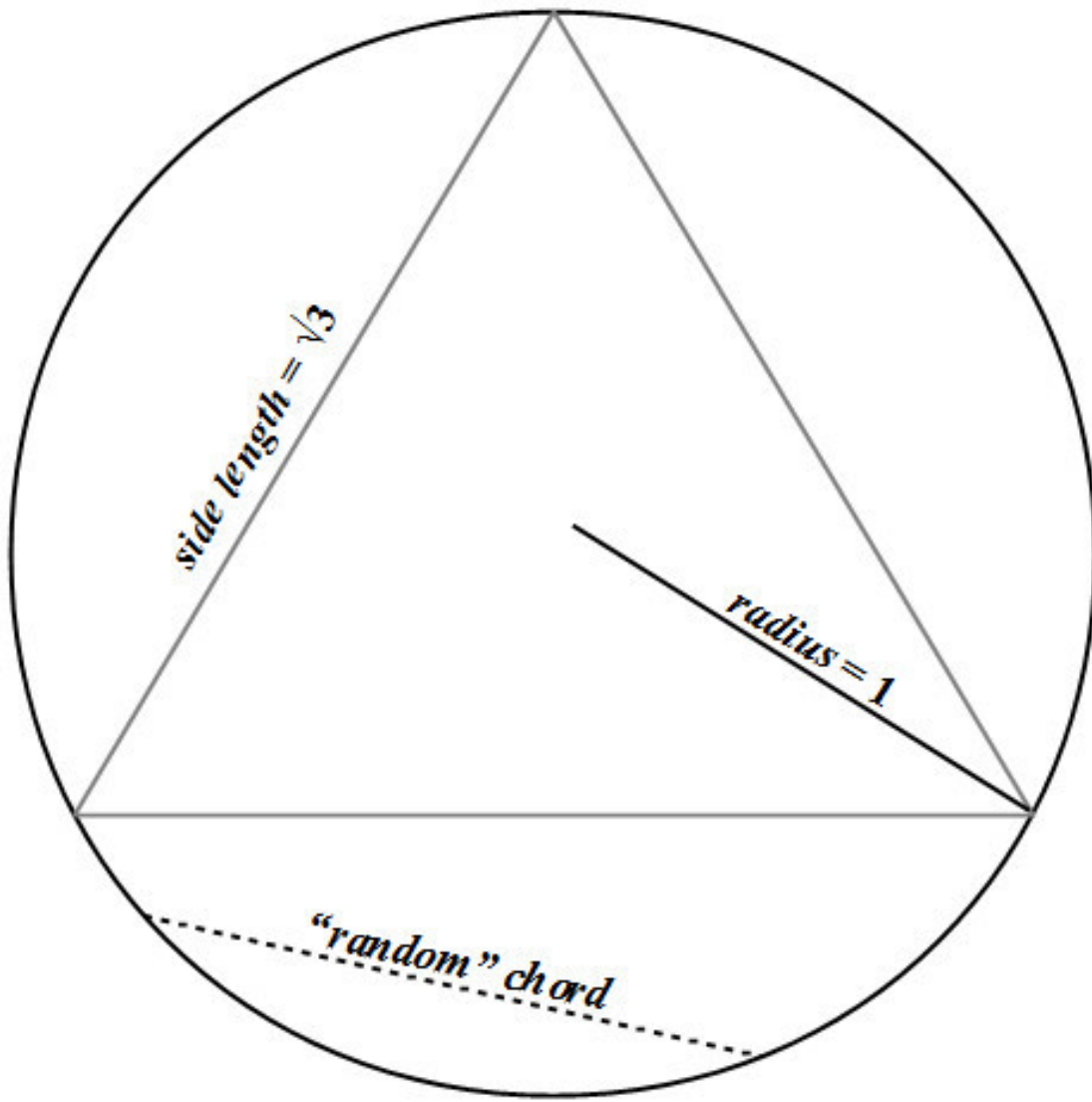
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31)Random Chord

Recall that a *chord* is a line segment between two points on a circle. If the two points on the circle are close to each other, the chord will be very small. If the two points are as far away as possible, then the chord is known as a diameter, the longest chord.

The random chord problem is the following. Let's say you draw *random* chord on a circle. The chord could be extremely small, or it could be as large as the diameter. The question is, what is the probability the chord is longer than the side of an inscribed equilateral triangle?

(Alternately stated: On a circle with radius 1, what is the chance that a random chord has length longer than $\sqrt{3}$?)



This type of problem is an example of *geometric probability*. The technique to solve is to (1) figure out the range of sizes for the random chord, and then (2) figure out the proportion of sizes would make the required condition true.

For instance, if we (1) found the random chord varied in length from 0 to 2 with equal chance, and we (2) wanted to know when length was larger than 1, then we would say there is a 50 percent chance the length will be larger and a 50 percent chance it will be smaller.

What is the answer to the random chord problem?

A tale of three answers

Imagine a math teacher assigned this problem. The next day Alice, Bob, and Charlie all return to the classroom with their answers. Alice confidently says the answer is $1/3$. But Bob protests and claims that he has a proof showing the answer is $1/2$. Just as the class starts to take sides, Charlie says they are both wrong. He believes the correct answer is $1/4$, and he has a proof to back up his claim.

The argument was getting pretty intense (for math class anyway), and so the students ask the teacher to provide the correct answer.

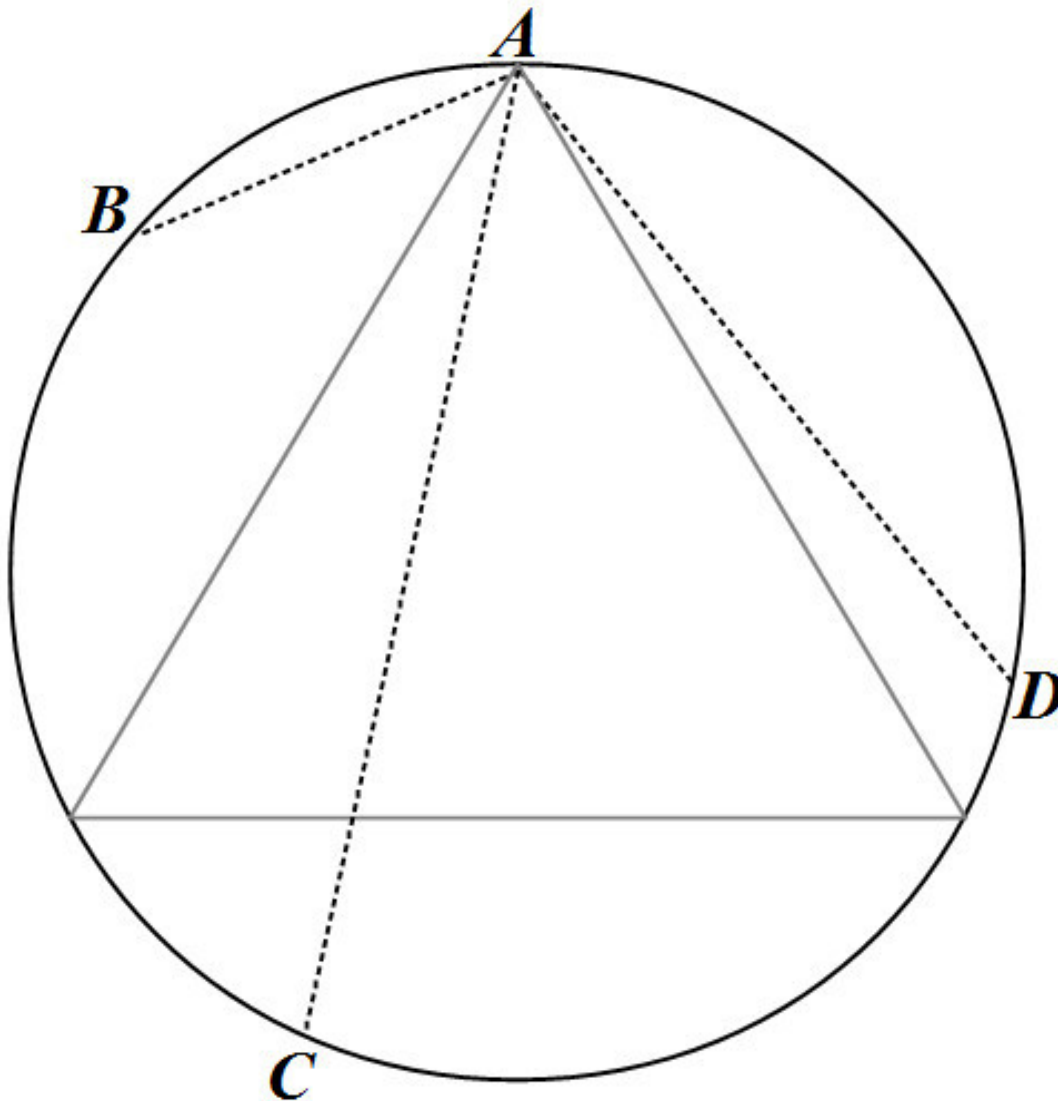
The teacher then says the most puzzling of things. He says that Alice, Bob, and Charlie are all correct. The problem, he explains, was a trick. The question is not what the correct answer is. The question is, how do you define a *random* chord?

Alice's attempt: pick 2 random points

A chord is a line segment between two points on a circle. Alice interpreted that a “random” chord would be a line segment that connects two random points on the circle. If the two points happen to be close to each other, the chord will be small, and if the two points happen to be far apart, the chord will be large.

What are the chances that two random points are far apart so that the chord is longer than the side of the triangle?

We don't need to do too much computation if we utilize a good trick. Let's imagine that we have picked two points on the circle. Now, rotate the equilateral triangle so that one of its vertices coincides with one of the endpoints of the chord. This won't affect the calculation; it will just make it easier to visualize the answer. In the following figure, the point A is chosen to coincide with a vertex of the triangle.



Compared to the side of the triangle, the chords AB and AD are smaller while the chord AC is larger

The second random point for the chord can be in any of the three regions (the triangle endpoints are the boundary points for three arcs containing points B , C , and D). Each of these arcs is equal because the equilateral triangle divides the circle into three arcs of equal length. So it's equally likely that the random point could either be in the same region as B , C , or D .

In two of those arcs—the ones that contain points B and D —the chord will be smaller than the side of the triangle.

It is only in the third case—when the point is in the arc containing point C —that the chord will be longer than the side of the triangle.

Therefore, we conclude there is a $1/3$ chance that a chord will be larger than the side of a triangle, if we pick two random points on the circle.

Alice felt this was the correct answer. But in fact, it was just one way to interpret the problem, and Bob thought up another equally justifiable ways to pick a random chord.

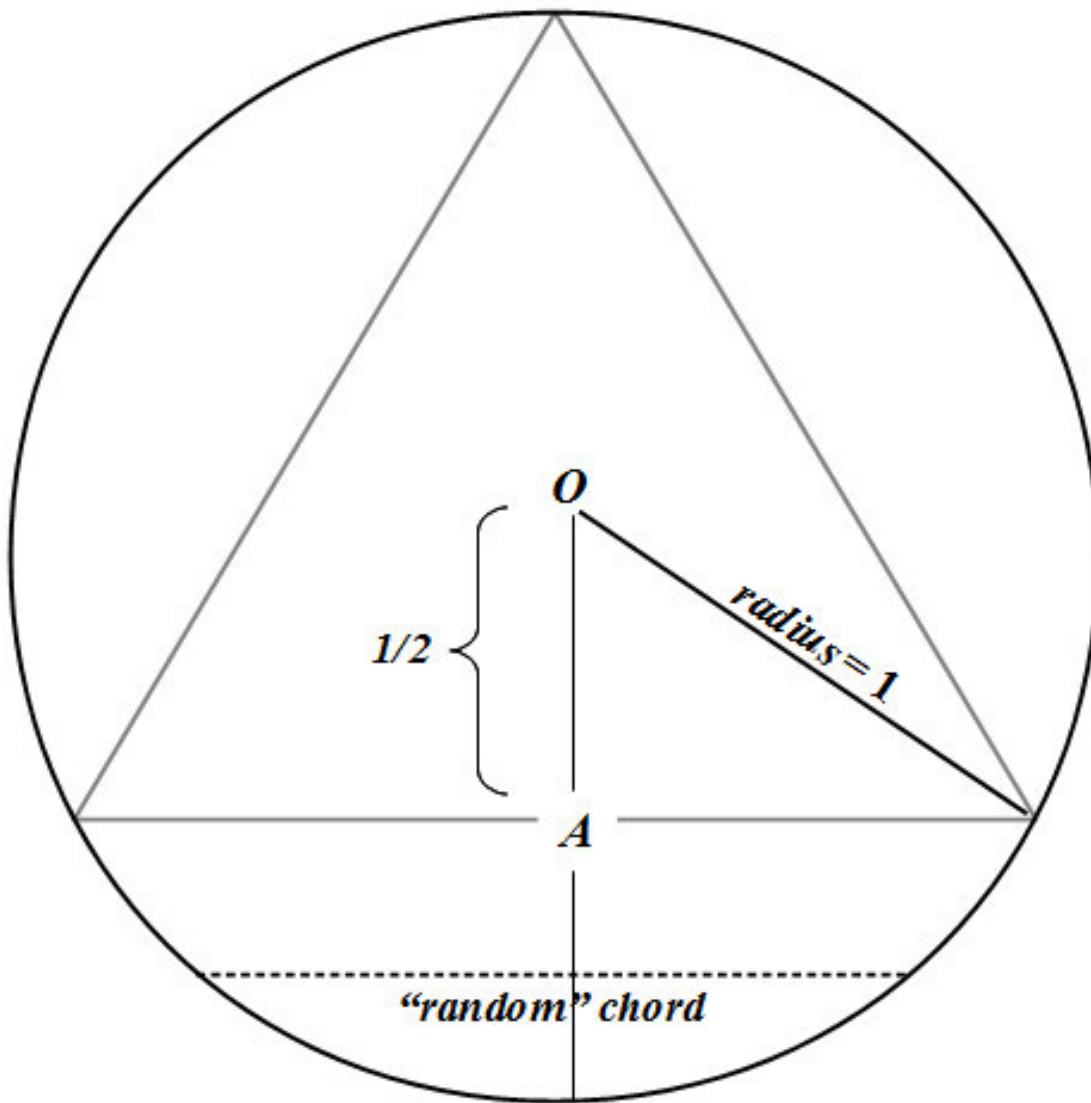
Bob's method: random radius

Bob did not think of a chord in terms of two points on a circle. He likes to define things on a circle in terms of the radius. He chose a random chord in the following steps. The first step was to draw a *random* radius on the circle. The second step was to choose a *random* point on this random radius. The third step was to draw a perpendicular line to the radius at this point. The resulting chord was Bob's definition of a random chord.

Now this all might seem a bit complicated. Why would Bob pick a random radius and a random point? Why can't he just agree that Alice's method makes more sense?

Part of the reason is mathematical exploration. It is sometimes useful to tackle a problem from a variety of methods. The processes might produce the same result—which means one can choose how to solve the problem. Or it might be that the methods are incompatible and offer different results. There is nothing wrong with Bob's definition of a random chord, so we need to figure out the answer that it yields .

Let's use Bob's method and calculate the probability. We will first pick a random radius. Then, we will use the trick of rotating the triangle to make one of the sides perpendicular to this radius, as in the following figure (remember, this doesn't change the calculation, it just makes it easier to visualize).



A random chord from a point below A will be smaller than the side of the triangle; a chord from a point above A will be larger

Point O is the center of the circle, and point A is where the radius intersects with the side of the triangle. It is easy to verify point A is the midpoint of the radius. One possible chord is drawn from a point that is below point A . From comparison it is clear this chord is smaller than the side of the triangle.

The random points can be chosen anywhere along the radius of length 1. From the figure, it is easy to deduce that if the random point chosen is below the halfway point A , then the resulting chord will be smaller than the side of the triangle. Similarly, if the second point is above the halfway point A , and closer to the center, then the resulting chord will be larger than the side of the triangle.

Therefore, half of the random points will result in a chord longer than the side of the triangle. So Bob says the answer should be $1/2$, which is different from Alice's answer of $1/3$.

Alice and Bob eventually understand they have to “agree to disagree” in that each sees how the other solution makes sense. Charlie is the one who throws a wrench in the picture because he offers a third, equally justifiable way, of looking at the problem.

Charlie's method: random midpoint for chord

Recall Bob's method for a moment. Bob constructs a random chord by: (1) drawing a random radius, (2) picking a random point on the radius, and (3) drawing a perpendicular to the radius at the point.

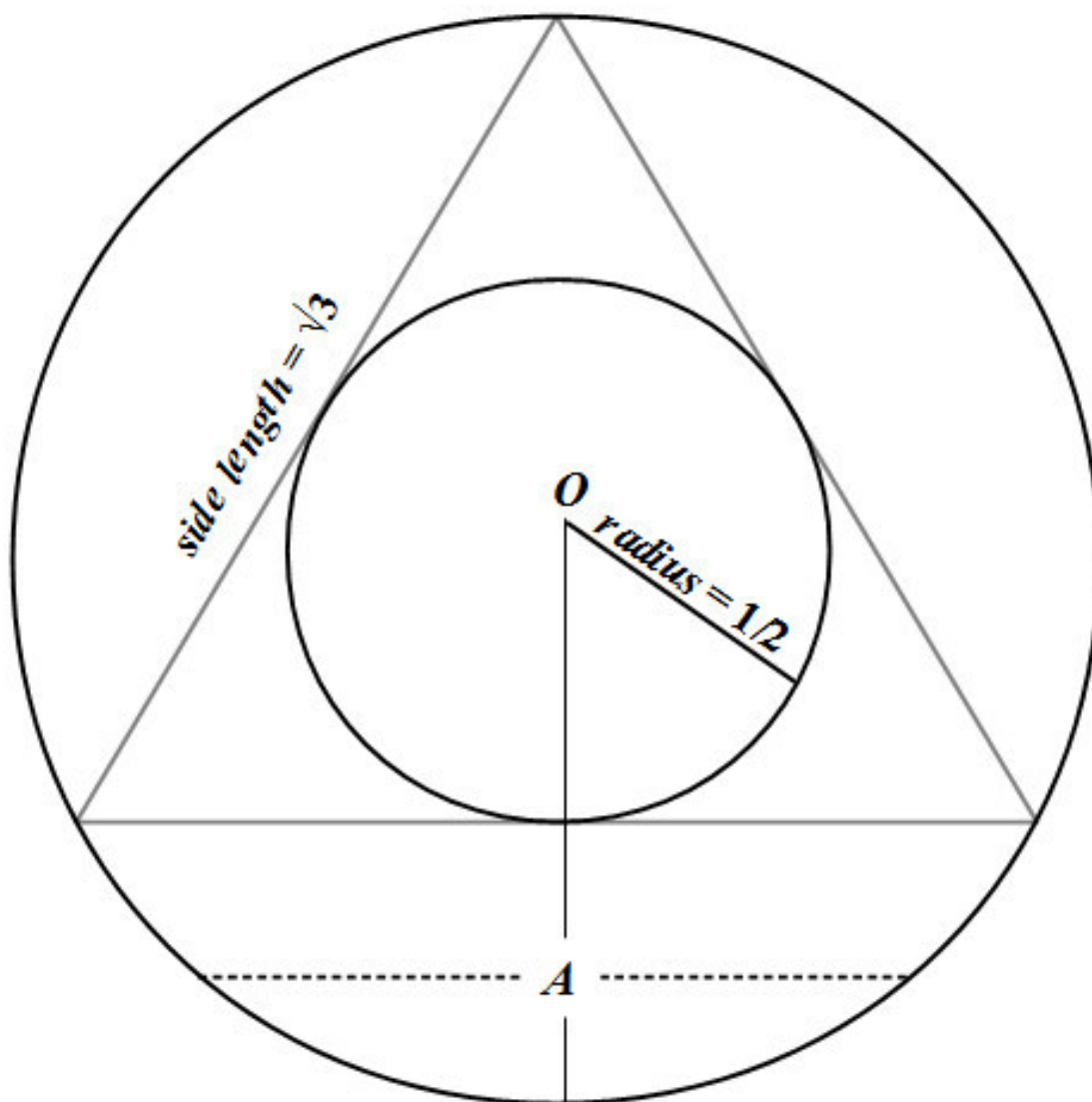
Charlie came up with pretty much the same construction, except he reversed the order of steps (1) and (2). He begins by (1) picking a random point on the circle, anywhere but the center. Then he (2) draws a radius that goes through that point—there can only be one. Finally, he (3) draws a perpendicular chord to the radius at the point.

Why should it matter if you pick a random radius first or pick a random point? It might seem this small detail doesn't matter, but in fact it does change the answer by quite a bit!

(Charlie's method is also equivalent to picking a random midpoint for a chord. So the question is: for which points will the midpoint of a chord be larger and smaller than the side of the triangle?)

What is the answer? The trick is to realize the following. If the point is close to the center of the circle, the resulting chord will be large and almost

the size of the diameter. If the point is far from the center, then the resulting chord will be small. What is the boundary point when the chord will be exactly the same size as the side of the triangle? The following figure can assist in this calculation.



If the point A is outside of the circle, then the resulting chord will be smaller than the length of the triangle side $\sqrt{3}$

In the above figure, there is a circle drawn inside the triangle. If the random point is exactly on the boundary of this circle, then the resulting chord will have the same length as the side of the triangle (to see why, note that the three sides of the triangle are tangent to the circle).

Now we can reason the chord will be *longer* than the side of the triangle if the point lies *inside* the inscribed circle and will be *smaller* if on the *outside* .

The probability a point in inside the inscribed circle is the ratio of the area of the smaller circle to the larger one. Basic geometry can show the inscribed circle has a radius that is half as big as the original circle (in the figure above, the radius is $1/2$ compared to the big circle of 1). Hence, the ratio of the areas is $1/4$.

Therefore, there is a $1/4$ chance the random chord will be longer than the side of the triangle.

Ultimately, Alice, Bob, and Charlie all agree to see the light in each other's methods. The heart of the problem is not to find a single mathematical answer. But it is to realize that one needs to be very careful when posing a question about randomness, as sometimes equally justifiable methods result in different answers.

GAME THEORY PARADOXES

What's the right decision to make? The answer depends not only on what you know and how you act, but also on what other people know and how they will act *and respond* . Game theory has the lofty goal of studying such interactive decision-making amongst competitive, rational agents. Because game theory is built upon logic and probability—both areas that are filled with paradoxes—it is no surprise that the study of strategic thought has its fair share of paradoxes.

Rationality is a tricky concept to define, and that makes it the subject of several paradoxes. One issue arises in games where the payoff depends on the intent of the player. In the first paradox, *You Can't Choose Wrong* , there is a conflict between making the ostensibly rational choice which ends up with a lower payoff than the presumptive irrational choice.

Another issue is that playing the irrational choice may provide a strategic advantage, which again makes the irrational choice feel like the rational thing to do. This is the nature of the *Centipede Game* and the remaining paradoxes.

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32) You Can't Choose Wrong Paradox

You're at a job interview and you are requested to make a simple decision¹.

The interviewer says, "I have a \$1 bill and a \$5 bill, which one would you like? Don't worry, if you make the wrong choice, I'll give you an extra \$5. What is your choice?"

You think carefully about the problem. It looks like the interviewer is trying to check if you can reason out a little bit. You think, only a fool would take the \$5 and be greedy. The smart choice must be to take the smaller \$1 bill. Then the interviewer would inform you that you made the wrong choice, and he'd give you an extra \$5. In the end, you'll have \$6 and more money than if you took the \$5 bill.

But just as you're about to indicate your answer, a thought strikes you. If the \$1 bill ultimately nets more money, wouldn't that make the \$5 bill the wrong choice? Using that logic, you realize you should probably take the \$5 bill.

Now it gets interesting. If you think taking the \$5 bill is the right choice, then that means taking the \$1 bill was the wrong choice, and hence your first instinct was right!

The paradox is that no matter which choice you make, you will end up making the wrong choice:

–if you think the \$1 bill is the right choice, then the \$5 bill is the wrong choice and would pay out \$10

–if you think the \$5 bill is the right choice, then the \$1 bill is the wrong choice and would pay out \$6

An economist might go even further. The *principle of revealed preference* is that if one is given a choice of A and B, and one actually chooses B, the action *reveals* that that choice B was preferred over choice A. After all, why would someone make the bad choice on purpose ?

Applied to this situation, any choice you make must be preferred to the choice you did not make, so no matter what you do, the other choice is the wrong choice. You'll never end up getting the extra \$5 payout.

You realize this and indicate the paradox to the interviewer. You figure that you cannot count on the “wrong choice bonus money,” so your safest bet is to take the \$5 bill. Besides, it's safer to cash out with \$5 for sure, with the possibility of winning \$10, then to try and gamble on the \$1 choice for a maximum \$6 payout.

The interviewer praises your answer. He tells you he has a way of scoring interview answers as follows:

0 points: picking \$1 with no justification

1 point: picking \$5 with no justification

2 points: picking \$1 and saying why it's the “wrong” choice

3 points: picking \$5 and explaining the second-order reasoning for why it's the wrong choice

4 points: explaining the paradox

5 points: additionally offering an analysis of risk and taking the \$5

The game is analogous to a dilemma for parents raising children. Parents tell their kids about good and bad choices, and eventually the kids grow up and have to choose on their own. They tell the child to make the right choice, but if the child messes up, the loving parents will be supportive and help the kids recover.

A naughty child might purposefully misbehave (akin to taking the \$1 bill), knowing he will get bailed out and essentially be rewarded for making the wrong choice. In this game the hard choice lies with the parents as they balance support and safety against the risk of encouraging bad behavior.

A good boy may realize the range of outcomes, from doing the right choice to occasionally making the wrong choice to get attention and test the limits.

Ultimately he might realize the tension and make the smart choice for himself. After all, making the right choice (akin to taking the \$5 with a risk analysis) is the highest guaranteed payout, and the smart boy is not going to gamble with his future.

(As a final note on the original game, one should note there is no paradox if the game is slightly rephrased. If the game is, “I will pay you an extra \$5 if you make the *right* choice,” the only correct answer will be to take the \$5 and get rewarded. Positive reinforcement for the win...)

Notes

1. This paradox is adapted from the “unachievable wisdom paradox” listed in Nicholas Rescher's *Paradoxes: Their Roots, Range and Resolution* (Open Court, 2001).

33)Parrondo's Paradox

Consider a gambler who visits a casino. One night he decides to play game A several thousands of times. Statistically, he is playing a game where the odds are rigged against him, and he is likely to lose money. The next night he plays game B, which again, is designed to be an unfavorable bet.

Now imagine that a third night he plays game A and game B alternately. What do you think will happen?

If the two games were slot machines, the gambler would again expect to lose money. Playing two unfavorable games should be not better than playing each unfavorable game, right?

Remarkably, that is not always the case! It is possible to create two games A and B which are each individually losing bets, but played one after the other can be a winning bet. This phenomenon is known as Parrondo's paradox, first described¹ by Professor Parrondo in 1996.

How is it possible that two losing bets can combine to create a winning bet?

The trick is the type of game that is being played.

A standard game

Let's play a simple game. I'll flip a coin. If it's heads I'll pay you \$1, and if it's tails you pay me \$1. This is a game that is fair to both parties: you and I each have even odds of winning, and the payouts are even. The expected value to the game is:

$$E(\text{game}) = \text{Pr}(\text{win})\text{gain} + \text{Pr}(\text{lose})\text{loss} = 0.5(1) + 0.5(-1) = 0$$

Now let's say that I rig the coin so that it comes up tails 51 percent of the time. Clearly I will have an edge and the game is unfavorable for you, with an expected value of less than 0:

$$E(\text{game}) = \text{Pr}(\text{win})\text{gain} + \text{Pr}(\text{lose})\text{loss} = 0.49(1) + 0.51(-1) = -0.02$$

You can expect to lose 2 cents each time you play this game. And over time, you will lose money and I will gain money.

If game A is a biased coin where you lose 2 cents, and game B is a biased coin where you lose 3 cents, you will lose in each individual game. And you will also lose when you combine the two games in any playing order. There's no way you can beat a casino by simply playing the right slot machines in some particular order.

These types of games depend only on the payouts and whether you get a lucky streak in the short-run. They will always be losing in the long-run.

Parrondo's paradox depends on a special type of game which varies based on how much money you have on hand. The game can change depending on whether you have \$1 or \$5 on hand. These are known as *capital dependent games*.

A capital dependent-game

Now consider the game that involves two coins. The game is if you have an even amount of money on hand, we will flip a coin that offers you a 0.09 chance of winning. If you have an odd amount of money, then you will flip a coin that offers a 0.74 chance of winning.

This is not like a typical casino game: each time you play, you will either gain or lose 1 dollar, which changes how much money you have, so you flip the two coins on alternate turns.

Parrondo's paradox depends on these types of games.

A very simple example

Let's say that you play the following game A. The rules are simple: you lose \$2 every time. This is clearly a losing game.

In game B, you will gain \$5 if you start with money that is a multiple of 3, and you lose \$3 if you have money that is not a multiple of 3. On average, if you start out with a random amount of money, Game B is also losing.

Roughly $2/3$ of the time you will lose \$3 and only $1/3$ of the time you gain \$5.

Let us verify that both games are losing if you start out with \$5 .

In game A, if you start with \$5, you lose \$2 each turn to get \$3, \$1, and then you go bankrupt.

In game B, you will lose \$3 to be at \$2, and on the next turn you go bankrupt.

So each game is losing if played individually.

But consider if you start with \$5 and play games A and B alternately. First you lose \$2 in game A which puts you at \$3. Then you play game B and you gain \$5 to end up at \$8. Then you play game A and you lose \$2, which puts you at \$6. Then you play game B and you go up to \$11. After playing games A and B, you end up gaining \$3 on each round. That is, the two games played alternately is a winning bet!

And that's the trick of Parrondo's paradox² . While each individual game is a losing bet, the two played in some order can create a winning bet.

A more precise explanation is the following. Game A is a single bet while game B really includes two bets. The bet in game A and one of the bets in game B are losing bets. The other bet in game B (to earn \$5) is a very favorable gamble. Under normal circumstances, playing game A or playing game B alone is a losing bet. But when the two games are played in sequence, it is possible the results from game A allow the gambler to play the favorable bet in game B. For that reason, playing the two games could result in a winning bet.

Parrondo's original example³ is more nuanced than that. The games in the situation are:

Game A: biased coin that loses just over 50 percent of time

Game B: one coin that loses just over 90 percent of time—you play if your capital is a multiple of 3, other coin that wins just under 75 percent of time—you play if capital is not a multiple of 3

Clearly game A is a losing one. Showing game B is a losing game is a bit more complicated. The proof depends on analyzing the various “states” that a gambler can be in (dependent on his money), which game he plays, and the long-run average. The process to solve this game is known as a static Markov chain analysis. One can show the long-run average is negative and the game is losing (roughly speaking the capital is a multiple of 3 in 1/3 of cases, with a 90 percent loss rate, plus it's not a multiple of 3 in 2/3 of cases, with a 3/4 chance of winning. This roughly nets to a slightly negative expected payout.)

Playing each game individually results in individual losing bets. But if you play AA and BB alternately, a neat thing happens where you end up winning money on average. Proving this is a bit complicated, but the idea is the same as the simple game presented above: playing the two games gives a favorable distribution of playing the coin with high odds of winning. (See a simulation here⁴ : <http://bit.ly/XZ8msR>)

The devious part is that we can also reverse the logic. We could also construct two games that were individually winning but when played one after another the two would be a losing bet. But that sounds a lot more depressing so Professor Parrondo chose the right example.

Notes

1. The paradox has an official website with its history
<http://www.eleceng.adelaide.edu.au/Groups/parrondo/intro.html#introduction>
2. The simple example is adapted from an example on Wikipedia
http://en.wikipedia.org/wiki/Parrondo's_paradox#A_simplified_example
3. Same as 1.

4. The full web address for the simulation is <http://jean-paul.davalan.pagesperso-orange.fr/proba/parr/index-en.html>

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34) Centipede Game

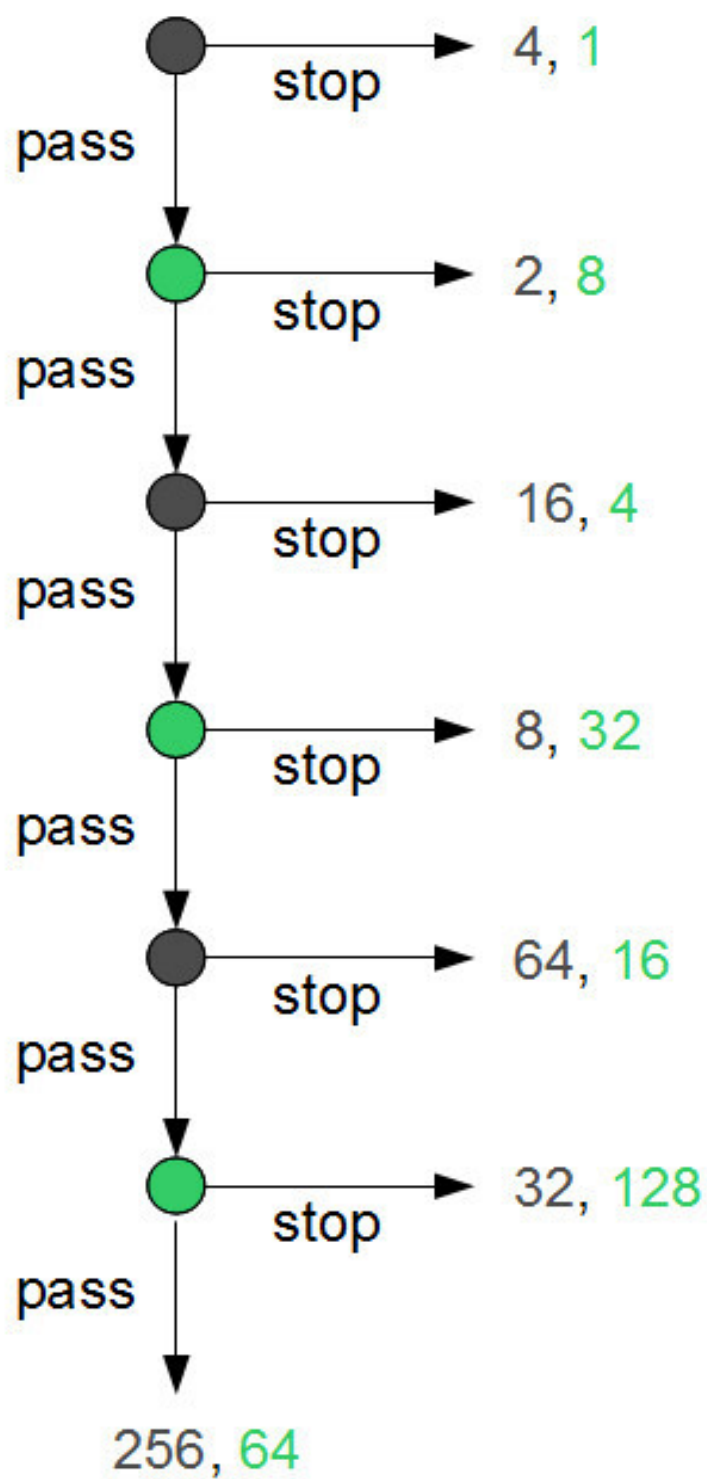
Suppose I'm sitting from you in a laboratory experiment. The table has two piles of money: one with \$4 and another with \$1.

You go first. You either take the \$4, ending the game and leaving the \$1 for me, OR you pass.

If you pass, the experimenter will come in and double the amount of money in both piles. Then, it's my turn. I can either take the larger pile of \$8, ending the game and leaving \$2 for you, or I can pass, knowing the experimenter will double both piles of money and the game goes to your turn.

We both know the game continues for up to 6 rounds, as illustrated in the following diagram. The payoffs are listed for (first player, second player) and the two players alternate who moves, with the first player moving first.

A CENTIPEDE GAME



How would you play this game? Do you take the money, or do you play it out for a few rounds ?

The paradox of trust

The game is named the centipede game because the game tree resembles the many-legged insect. The unselfish move is to pass and let stakes grow. After all, when money is increased, there is more for everyone in society to share, and theoretically everyone can be better off.

But the money is not divided evenly, and that's what drives the selfish move to take. In any given round, a player is entitled to take the larger share of money. A player that passes gives the other player the opportunity to grab the larger pile. The game is designed so a player that passes ends up with less money if the next player ends in the game in the next round. Or, equivalently, a player can guarantee a larger payoff by grabbing the pile now over ending up with the smaller pile in the next round.

In any given round, each player is tempted to take the larger pile. But if both can trust each other, the piles can grow and both can be better off. Each player has to trust the other will also grow the pile, but both know the game will ultimately end where some player short-changes the other.

The Nash equilibrium offers a unique prediction: the first player will grab the larger pile right away. The reason is neither player can trust the other player will grow the piles, and so each player opts for the larger payoff immediately.

But this is not how most people would play the game.

Experimental results

Early research illustrated divergence from the game theory prediction. Less than 1.5 percent of games conducted in one experiment showed people stopping the game in the first round. Worse, this result held true even after people repeatedly played the game, demonstrating that experience and the effect of learning did not nudge people to the Nash equilibrium.

It did not matter if researchers added more rounds of play, or if the stakes were increased. Overall, the Nash equilibrium was still not played by at least 25 percent of participants .

But a more recent study did find people exhibiting strategic play. The twist was the participants were chess players, who are used to strategically reasoning several moves in advance.

When the centipede game was played with chess players, nearly 69 percent ended the game in the first round. Furthermore, the results showed a larger convergence to the Nash equilibrium for players of higher skill rankings. Here are statistics of games that ended in the first round by increasing skill level:

Lowest rank: 61 percent

Federation masters: 73 percent

International masters: 76 percent

Grandmasters: 100 percent

A remarkable 100 percent of chess grandmasters played the Nash equilibrium according to backwards induction.

It is important to note that neither player knew the skill level of the other. In the field experiment, people were put into different rooms and told they were playing another "player." It would be fair to say they knew they were playing another chess player, but they were not aware of the other person's skill.

The results of these field experiments were confirmed in a laboratory setting. In general, chess players were able to learn that if their opponents were using backwards induction (they were also a chess player), then they would end the game earlier on as well.

While the experiment illustrates people can reason strategically, it does not solve the paradox of the centipede game that both players would be better if they could trust each other for a few rounds. The people who "irrationally"

pass to grow the pot are the ones who end up with more money than the people who "rationally" opt for the payout in an early round.

Note

1. Ignacio Palacios-Huerta & Oscar Volij, "[Field Centipedes](#) ."

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35)Hypergame Paradox

Alice and Bob love to play board games and one day decided to catalog all of them. One by one they would write the name of each game they owned—checkers, tic-tac-toe, chess, etc.—each game on its own index card.

But quickly they ran out of games. They decided to start writing the names of games they didn't own (which they could go and purchase). And when that wasn't enough, they included mathematical games.

The only rule was they had to include games that were like actual board games: that is, the players moved alternately in turn, at each move there is a finite number of possible actions, the game would end after a finite number of moves, and there were no ties (in a draw, the player who moved first was declared the winner).

Being mathematicians, they were curious about the theoretical nature of their exercise. How large was the set of all games they could write? They suspected they could write infinitely many such games. But they were in for a bit of a shock.

An impossible game

Alice came up with the following thought experiment. Suppose she had a box that contained all possible games—call that set G .

She now dreams up the rules for another game, called the hypergame, that she writes on a card. The game is this: the first person to move calls out a game from the permissible games G , and then they play that game. You can think about hypergame like dealers-choice poker, in which the rotating dealer gets to pick which version of poker is played.

Alice then faces a cataloging decision: does the card for hypergame belong inside the box or not?

On the one hand, the card clearly should belong to the box. Hypergame is a game where players move alternately, at each move there is a finite number

of possible actions, and the hypergame should end in a finite number of moves (one more move than the board game they play) .

But if she puts the card into the box, then that would mean the hypergame was an allowable choice for a game to play.

In many instances, playing hypergame might go smoothly: Alice might name out to play tic-tac-toe, they play it, and it's all over.

But hypergame could also end up in a terrible gridlock. How is that? Let's say that Alice calls out to play hypergame. Now they are playing hypergame. Now Bob, who plays the first move in hypergame, gets to choose which game *he* wants to play. He could very well call out to play hypergame. That sends the move back to Alice, and on *her* first move, she could also call out to play hypergame. By the rules of hypergame, each can keep legally calling out to play hypergame.

Now for the catch: since Alice and Bob can play hypergame infinitely with no end, this means hypergame is not a finite game. Therefore, the card with hypergame has no business being in the box that contains only finite games.

In other words, hypergame should clearly be included in the set of finite G , but if it's part of the set G , then clearly it is not a finite game and should not be part of G .

So what's the answer: what kind of game is hypergame?

An explanation, of sorts

The hypergame paradox is caused by defining a game in which you are allowed to play the game as the first move. This is a self-referential paradox, along the lines of Russell's set paradox or the liar paradox discussed earlier. Modern set theory resolves the paradox by saying such sets are not properly defined. In other words, the hypergame is not a properly defined game.

But of course you protest: you know it's a game since you can play it!

Notes

1. [Playing Games with Games: The Hypergame Paradox.](#) William S. Zwicker, The American Mathematical Monthly. Vol. 94, No. 6 (Jun. - Jul., 1987), pp. 507-514.

[OceanofPDF.com](#)

36)The Wallet Paradox

Bill Gates meets Warren Buffett at a dinner party and the host tells them to play a game. Each person will place his wallet on the table. The person with less money in his wallet wins all the money.

Is anyone favored to win this game? What happens when the game is repeated?

A matter of perspective

At first glance, the game seems favorable for Bill Gates. (The initial logic and first two resolutions are along the lines of the necktie paradox, repeated here for completeness).

He might think as follows. He could either have more or less money, so he has a 50/50 chance of winning the game. If he loses, then he would lose only the money in his wallet. If he wins, however, then he would win more money than is in his wallet.

That is, Bill Gates can wager the money in his wallet (say x dollars) and have a 50 percent chance of winning the money in Warren Buffett's wallet, which is more than x dollars. This is obviously a favorable gamble, and thus Bill Gates will want to play.

On the other hand, Warren Buffett thinks about the game. By symmetry, he can reason exactly the same thing. He feels that he has an even chance of winning more money than he has in his wallet, and thus he is favored by the game.

Both billionaires find the game advantageous. And yet this cannot be. The game is a zero-sum game (the profits of one are exactly the losses of another) and thus it is impossible for both to expect a profit at the same time.

How can the apparent paradox be resolved?

Resolution 1: fix the reasoning

Bill Gates and Warren Buffett are unlikely to reason so wrongly. They would see the error in the above logic.

The problem with the reasoning is that each is considering his wallet as being both the one with more money and less money at the same time. In reality, a wallet can either be the one with more money or it can be the one with less money.

The correct logic is:

–If I have the wallet with more money, and I play the game, then I will lose my more valuable contents

–If I have the wallet with less money, and I play the game, then I will win the other wallet that is more valuable

There is an equal chance of being in either situation. Therefore, the game does even out as either player can win or lose the wallet with more money. The game will average out to Bill Gates winning half the time, and Warren Buffett winning half the time. Thus neither is favored and the game is fair.

Resolution 2: make a proper expectation

Another way to see the game is fair is to write out the math carefully. Suppose the more valuable wallet is worth x dollars more.

A player either wins x if he originally had less money, or he loses x if he had more money. So the expectation is:

$$E(\text{bet}) = \text{Pr}(\text{win})\text{gain} + \text{Pr}(\text{lose})\text{loss} = 0.5(x) + 0.5(-x) = 0$$

The math demonstrates the game has a zero expected payoff to either player, and thus the game is fair.

Repeated play

Afterward, it turns out both Bill Gates and Warren Buffett loved the game.

They add a twist as follows. They will play the game every day for a month. To manage the stakes, they agree to carry an average of \$100. Is this game fair? What is the best possible strategy, given you know what your opponent is doing?

Winning strategy in repeated play

The game gets more interesting when repeated. Even when players agree to carry the same amount of money on average, the game is not necessarily fair and one player can be favored.

Consider the example of Bill Gates and Warren Buffett agreeing to carry \$100 on average. Suppose they limit their strategies to the following three:

Strategy A : carry \$100 every time

Strategy B : carry \$75 half the time, and \$125 half the time

Strategy C : carry \$75 two-thirds of the time, and \$150 one-third of the time

We can evaluate how the strategies fare against each other.

It turns out there is a rock-paper-scissors type of situation: strategy A is profitable against B, which is profitable against C. But then strategy C is profitable against A!

This cyclical nature gives a hint about the nature of the game, more on that in a bit. First, here is the math that confirms it.

Strategy A vs. B

For shorthand, write (x, y) as the amounts in each of the two wallets.

Strategy A is to carry \$100 always. Strategy B is to carry either \$75 or \$125.

The strategies will lead to the cases $(100, 75)$ and $(100, 125)$, each happening with equal probability.

In the (100, 75) case, the person employing strategy A loses the entire \$100.

In the (100, 125) case, the person employing strategy A wins the entire amount for a profit of \$125.

The expectation of employing strategy A against B is thus:

$$0.5 (-\$100) + 0.5 (\$125) = \$12.5 > 0$$

This verifies that strategy A is profitable against strategy B.

Strategy B vs. C

Calculating this expectation is just a bit more involved, as both strategies are probabilistic.

There are four possible outcomes that can result. Here is how strategy B fares against strategy C.

(75, 75) → happens $(1/2)(2/3) = 2/6$, and is a draw

(75, 150) → happens $(1/2)(1/3) = 1/6$, and the player using strategy B profits \$150

(125, 75) → happens $(1/2)(2/3) = 2/6$, and the player using strategy B loses \$125

(125, 150) → happens $(1/2)(1/3) = 1/6$, and the player using strategy B profits \$150

Putting this together, the expectation is:

$$(2/6)(\$0) + (1/6)(\$150) + (2/6)(-\$125) + (1/6)(\$150) = \$8.33 > 0$$

Therefore, strategy B is profitable against strategy C.

Strategy A vs. C

There are two possible outcomes, each with different probabilities. Here is how strategy A fares against C.

(100, 75) → happens 2/3, and the player using strategy A loses \$100

(100, 150) → happens 1/3, and the player using strategy A profits \$150

Putting this together, the expectation is

$$(2/3)(-\$100) + (1/3)(\$150) = -\$16.67 < 0$$

Thus strategy A actually loses to strategy C.

In other words, we have verified that strategy A beats B, and strategy B beats C, and strategy C beats A, just like a game of rock-paper-scissors.

There is no one best strategy

The fact that strategy A is preferred to B, strategy B is preferred to C, and strategy C is preferred to A is a clue that the game might not have a single winning strategy. This hunch turns out to be true.

For any given strategy¹ with a fixed mean, one can find another strategy with the same mean that is profitable against it.

As a side point, this means the game has no Nash equilibrium: any strategy you dream up can be beaten, so you are never content sticking with a single strategy.

And I will close with something obvious but noteworthy. If someone asks you to play the game, you should definitely be skeptical. They probably have little or no money in their wallet and are hoping to make a quick buck.

It is only sensible to play the game in a fun party setting when both people are taken by surprise. It's even better after a night of drinking where both people are less likely to remember exactly how much is in their wallet.

Notes

1. There are some technical restrictions on what makes a well-defined strategy. The rock-paper-scissors example is adapted from the paper “The Wallet Paradox Revisited” by Maureen T. Carroll, Michael A. Jones, and Elyn K. Rykken. <http://academic.scranton.edu/faculty/carrollm1/wallet.pdf> . The paper also contains the proof there is no Nash equilibrium.

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37)Condorcet Paradox

This topic also appears in my book *The Joy of Game Theory: An Introduction to Strategic Thinking* .

In 1785, Marquis de Condorcet wrote one of his most important works, *Essay on the Application of Analysis to the Probability of Majority Decisions* . It was one of the first applications of probability to the social sciences, and it includes a stunning example of a possible problem with elections.

Imagine that there are three candidates running for office. I'll call them A, B, and C.

In this concocted example, let's just suppose there are only three voters (or there are three political constituencies, all voting in party line).

Each voter has a preference over the three candidates. Let's say that 1 prefers A to B to C, 2 prefers B to C to A, and 3 prefers C to A to B.

I'll write the preferences in shorthand, with the most favored candidate first, and the least favored last.

Voter 1: $A > B > C$

Voter 2: $B > C > A$

Voter 3: $C > A > B$

Which candidate best represents the majority preference?

Let's consider putting A in power. The result would be great news for voter 1, but both voters 2 and 3 would be unhappy as their candidates lost.

Voter 3 counters that the election is unfair. Voter 3 argues that C is a much better politician than A. This is clearly a biased opinion because voter 3 wants C to win it all. But voter 3 also discovers that voter 2 is unhappy. And that's the secret to the paradox .

Voters 2 and 3 meet and realize that they both prefer C over A. Take a look at the preferences to verify.

Voter 1: $A > B > C$

Voter 2: $B > \mathbf{C} > \mathbf{A}$

Voter 3: $\mathbf{C} > \mathbf{A} > B$

Voter 3 has C as a first preference, and voter 2 has C as a second preference *over A*. This means both voters 2 and 3 would prefer C over A. So voters 2 and 3 decide that C would be a better person than A. They gang up and overrule voter 1, placing C in power.

The peace only lasts a few minutes because soon voter 2 has regrets about recommending C. Sure, C is better than A, but only by a little bit. In fact, voter 2 can only live with B being in power, his first preference. Is there a way to make that possible? Voter 2 decides to talk with voter 1, and soon learns something he likes. It turns out voter 1 also prefers B over C, as the preferences indicate.

Voter 1: $A > \mathbf{B} > \mathbf{C}$

Voter 2: $\mathbf{B} > \mathbf{C} > A$

Voter 3: $C > A > B$

So voters 1 and 2 now combine forces to decide that B would be a better person than C. Using the power of majority, they gang up and place B in power. I'm sure you're getting the picture by now. B is in power for only a very short time when voters 1 and 3 join forces because they both prefer A over B.

Voter 1: $\mathbf{A} > \mathbf{B} > C$

Voter 2: $B > C > A$

Voter 3: $C > \mathbf{A} > \mathbf{B}$

And once again, the case is made to put A in power.

Using majority preference, it turns out that A loses to C, C loses to B, B loses to A, and A loses to C again, and so on *ad infinitum* . This is an example of how individual rational preferences cannot always be aggregated neatly into societal preferences.

In this situation, no candidate can claim to be favored by the majority. It's impossible.

A related voting paradox

The Condorcet paradox is a theoretical example of how it can be impossible to aggregate individual rational preferences into a sensible decision. It is just one example of many issues with voting.

To illustrate another voting paradox, I have created a fictional voting scene that might have happened in the TV show *Mad Men* . The problem in this paradox is that voting on two issues separately can result in a different outcome from voting on both issues at the same time.

Here is the setup. You're Don, a partner in an advertising company. You uncover that your partner Lane has stolen company funds. The amount was relatively small, but it was theft nonetheless.

In the show, Don went with a gut decision. But let's imagine he had consulted the other partners in the company to be sure.

They agree to take a vote. They know Lane is valuable and want to be cautious. They will only fire Lane if they believe he acted maliciously and would repeat the crime again.

Secretly, this is what each believes:

	Malicious?	Repeat?
Don	Yes	Yes
Pete	Yes	Yes
Roger	No	Yes
Bert	Yes	No
Joan	No	No

Don calls for a show of hands. He asks, “Was the crime malicious?” and a majority of three people raise their hands. He then asks, “Would he repeat it?” and again a majority of three people raise their hands.

Don communicates the decision to Lane who protests vociferously that he was desperate and was not given fair compensation. Won’t they reconsider to give him another chance? Don says the decision is final, and Lane makes his final move.

Only too late, Don reflects on the matter. If he had simply asked “Should we fire Lane?”, then the conclusion would have been different. Only two of the people—Don and Pete—thought that Lane was both malicious and would repeat the crime.

Was Don wrong in what he did? Not necessarily. Don just did not realize that voting on separate issues can result in a different outcome from voting on the issues together. The issue is that one must be careful about exactly what is being voted upon.

38)Apportionment Paradox

This chapter also appears in my ebook *But I only got the soup!*

This is a mathematical article that demonstrates it is often not really possible to split the bill completely fairly.

Suppose three people Alice, Bob, and Charlie order a pizza for \$10. It turned out that Alice and Bob were very hungry, eating 6 slices a piece, and Charlie just had 2 slices.

They want to split the pizza cost fairly, but they are subject to a practical issue. No one has coins to make change, so they have to round how much each owes to the nearest dollar.

What's the fair way to split the bill?

The highest remainder method

Here's a table that shows the exact amount each person owes based on the number of slices eaten.

PIZZA COSTS \$10

	Slices	Fair share
Alice	6	\$4.29
Bob	6	\$4.29
Charlie	2	\$1.43
Total	14	\$10.00

As mentioned above, no one has any change so the amounts have to be rounded. They agree to the following popular method (previously used in Congress to determine seats) known as the highest-remainder method: "Each person's fair share is **rounded down** to the nearest dollar, resulting in a shortage of money. The shortage is met by assigning dollars, one by one,

to the person or persons whose fair shares were rounded down the most (that is, the decimal portion of the fair share was the highest). ”

In the pizza example, the amounts are rounded down as follows:

PIZZA COSTS \$10

	Slices	Fair share	Rounded
Alice	6	\$4.29	\$4.00
Bob	6	\$4.29	\$4.00
Charlie	2	\$1.43	\$1.00
Total	14	\$10.00	\$9.00

This division leads to a total of \$9, which is one short of the \$10 bill. Who has the chip in the extra dollar?

As seen in the table, Alice and Bob have remainders in their fair share of 0.29 whereas Charlie has a remainder of 0.43. Since Charlie has the highest remainder, he is on the hook for the extra dollar.

So the three divide the pizza as follows:

PIZZA COSTS \$10

	Slices	Fair share	Rounded	Re-rounded
Alice	6	\$4.29	\$4.00	\$4.00
Bob	6	\$4.29	\$4.00	\$4.00
Charlie	2	\$1.43	\$1.00	\$2.00
Total	14	\$10.00	\$9.00	\$10.00

The final result is that Alice and Bob owe \$4, and Charlie owes \$2.

But wait, isn't Charlie paying too much?

Charlie, being a cheap college student, is not happy with the outcome. He's paying a dollar per slice that he ate, and the other two are only paying 66 cents for each slice. He shouldn't be penalized for eating less.

Furthermore, Charlie, being mathematically inclined, points out something else. It just so happened the pizza total cost was \$10 and that meant he owed \$2 .

But if the pizza was actually \$11—an extra dollar—then Charlie would actually owe less. The same division method would calculate that he owes just \$1.

To verify this, let's go through the math.

Here is a table of the fair shares and *rounded down* values when the total cost is \$11, Alice and Bob eat 6 slices, and Charlie eats 2 slices.

PIZZA COSTS \$11

	Slices	Fair share	Rounded
Alice	6	\$4.71	\$4.00
Bob	6	\$4.71	\$4.00
Charlie	2	\$1.57	\$1.00
Total	14	\$11.00	\$9.00

The rounded down values are exactly the same as before (4, 4, 1), and the total amount is \$9. Now the group is short \$2 for their \$11 bill. Who has to chip in extra money?

In this example, Alice and Bob have remainders in their fair share of 0.71 whereas Charlie has a remainder of 0.57. Since Alice and Bob have the two highest remainders, this means they are on the hook for chipping in an extra dollar each.

Therefore, the division algorithm yields the following:

PIZZA COSTS \$11

	Slices	Fair share	Rounded	Re-rounded
Alice	6	\$4.71	\$4.00	\$5.00
Bob	6	\$4.71	\$4.00	\$5.00
Charlie	2	\$1.57	\$1.00	\$1.00
Total	14	\$11.00	\$9.00	\$11.00

Now Alice and Bob have to pay \$5 each and Bob pays just \$1 .

This is the strange result: *when the pizza costs \$1 more, Charlie is actually paying \$1 less .*

What is going on?

The Alabama paradox

This example is based on an actual historical example in America. In the mid-1800s, the seats of Congress were assigned by the highest-remainder method based on state population.

But in 1880 a strange thing was observed. When the seats of Congress increased from 299 to 300, Alabama actually decreased its representation¹ from 8 seats to 7 seats!

As a result, the Alabama paradox refers to any time that increasing the total seats can decrease one party's allotted shares.

The end result is that fair division of seats cannot be decided strictly on a mathematical basis, unless we are willing to put up with some anomalies in apportionment. One will need to consider other concepts of fairness when apportioning shares.

In the pizza example above, there are other ways to resolve the bill. The group could, for instance, flip a coin to see who has to chip in the extra money, or perhaps one person would just volunteer to throw in the extra dollar.

The lesson: fair division is almost never a simple mathematical calculation.

Notes

1. The historical numbers of how Alabama lost a seat can be found on the following website

<http://www.cvl.ua.edu/math103/apportionment/illALPa.htm>

[OceanofPDF.com](http://www.oceanofpdf.com)

39)Forward Induction

To begin, consider the following model of conflict. This game is one of the simplest examples in game theory, often the first example presented in a game theory course.

Alice and Bob are planning to go out for an evening. Bob wants to go to the football game, but Alice wants to go to the opera. Still, both would prefer to be with the other person than going out to an event by themselves.

Let's say each person gets 3 points to go to their favored event, 1 points to go to the other event but be with their significant other, and 0 points if the two do not go together, as in the following matrix.

(The pair (3, 1) means Alice gets 3 points and Bob 1 if they both choose going to the Opera (O). If they both went to Football (F), the pair (1, 3) reverses the payouts as football is Bob's preferred choice)

		Bob	
		O	F
Alice	O	3, 1	0, 0
	F	0, 0	1, 3

The question is: how will this game play out?

Solution 1: opera or football

This game is very easy to solve. We need to consider what each person's best response is, given what the other person might be doing.

Here is how Bob thinks about the problem:

–If Alice goes to the football game, then I should also go to get 3 points rather than going to the opera for 0 points

–If Alice goes to the opera, I might not like the opera, but if I go I'll at least get 1 point. If I watch football I would instead end up with 0 points.

Bob's strategy is to guess what Alice wants to do and follow.

Alice's reasoning is exactly the same: she wants to be with Bob, so she should pick the choice she expects him to do.

The outcomes that both go to football, or both go to opera are the solutions of the game.

These are the pure strategy Nash equilibria of the game.

The pleasant finding is that both players end up choosing mutually beneficial outcomes.

But the annoying part is the game has two different solutions. I mean which one do they end up doing? Do they go to football or opera? The concept of the Nash equilibrium falls short here in providing a specific prediction.

However, there are other ways to think about the game.

Solution 2: correlated equilibrium

If the game is repeated, as it would be in a relationship, there is a good compromise that Alice and Bob could agree to.

What they could do is flip a coin to decide whether they go to football or the opera. Over time the events will balance out, and they both guarantee that they are together. They will get an average payout of 2.

This seems like a very reasonable solution. It also makes a lot more sense they flip a single coin and coordinate their choices, rather than each player flipping a coin individually and hoping to end up at the same place (the

mixed strategy Nash equilibrium doesn't make much sense in this game: a good chunk of time the two end up in different places).

In a healthy relationship, there is compromise and people may choose the correlated equilibrium .

But a determined girlfriend (or boyfriend) would not tolerate this and would want to change the game.

(From here on, the example uses a determined girlfriend, but it could equally well be the boyfriend who is pushy.)

Solution 3: forward induction

Let's suppose Alice really, really wants to go to the opera, and she wants to convince Bob that it's in his best interest too.

Here is a drastic strategy that Alice could use to change the game.

Alice takes out a \$20 bill from her purse, and then announces the following: "Look Bob, I really want to go to the opera. And if I think there is even a chance we are *not* going, that would stress me out. I'll vent by burning this \$20 bill. I'm going to the room to think about what I'm going to do. Come knock on my door with your choice in one minute. But think about what's best for both of us."

Alice has introduced another strategy to the game called burning money. In this game, Alice first chooses whether to burn her own \$20 bill, and then both players choose where they would like to go.

Let's say that burning a \$20 bill will destroy 1 point of utility for Alice. How will this game play out?

This gets a bit complicated, but let me sketch out the idea.

The way to analyze this game is to write out a large matrix with all the choices and solve as one normally would.

Alice has two moves in her strategy: she can either burn the money or not, and she can either go to the opera or not. Let's abbreviate Alice's choice of "burn the money, go to opera" as *BO* , and similarly the rest of her choices as *BF*, *NO*, *NF*.

Bob also has two moves in his strategy. He needs to decide what he will do if Alice burns the money, and what he will do if Alice does not burn the money. Let's abbreviate Bob's choice of "if she burns the money I go to the opera, if she does not burn I go to football" as *OF* , and similarly the rest of his choices as *OO*, *FO*, *FF* .

The matrix of payouts is as follows:

		Bob			
		OO	FO	OF	FF
Alice	NO	3, 1	3, 1	0, 0	0, 0
	NF	0, 0	0, 0	1, 3	1, 3
	BO	2, 1	-1, 0	2, 1	-1, 0
	BF	-1, 0	0, 3	-1, 0	0, 3

How can we solve this game? The process is to consider which options are reasonable. The logic is several steps. First, Alice will inspect the payoffs and realize that *NO* is better for her than *BF* in every possible way. That means Alice will never play *BF* . Bob realizes this, and then he considers

what the game looks like at that stage. He will then figure out that playing *OO* is better than *OF* in every remaining possible outcome.

This process of repeatedly removing bad strategies is how the game is solved. The process continues until only one option remains reasonable: *NO* for Alice and *OO* for Bob.

		Bob			
		OO	FO	OF	FF
Alice	NO	3, 1	3, 1	0, 0	0, 0
	NF	0, 0	0, 0	1, 3	1, 3
	BO	2, 1	1, 0	2, 1	1, 0
	BF	1, 0	0, 3	1, 0	0, 3

That is, the equilibrium outcome is that Alice does not burn the money and goes to the opera, and Bob chooses to go to the opera whether Alice burns money or not.

That is, they end up going to the opera just like Alice wanted!

Here is the paradoxical part. Alice got her way because she threatened to burn money. But she never actually has to burn the money: she gets her way because she threatens to torch her own utility.

But in that case, how can the threat of doing something irrational be the rational thing to do?

The resolution is to admit that irrational can be beneficial and “rational.”

The model also seems to capture a little about how spoiled brats operate in real life. They do not always throw tantrums. They only have to threaten to throw a tantrum and act unhappy to force everyone into their choice.

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40)Braess Paradox

This paradox also appears in my book *The Joy of Game Theory: An Introduction to Strategic Thinking*.

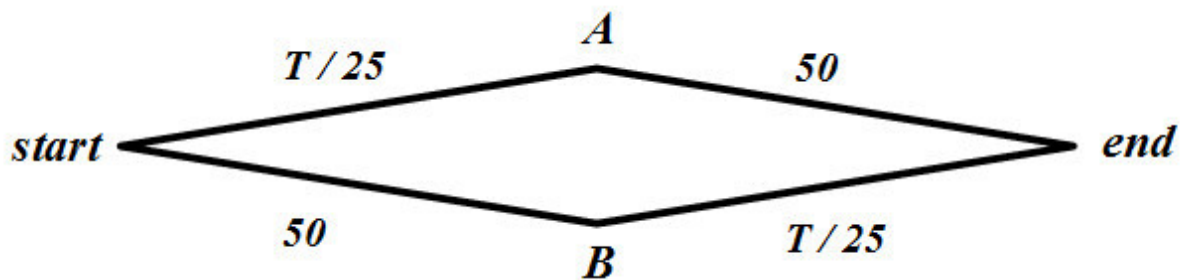
During the holiday season, several roads in my town were temporarily closed due to flooding. I was naturally worried about increased traffic. As it turned out, my fear was unfounded. In fact, I often found my travel times were reduced!

This experience made no sense to me. If traffic is about too many cars on the roads, how could blocking some roads speed up my commute? Or conversely, why might adding roads slow down my commute?

I did a little bit of research and came across an interesting game theoretic explanation known as the Braess paradox. It states that it's possible that adding a road could lead to slower travel for all drivers. Let's go through an example to see why.

A traffic question

Consider a traffic network where 1,000 drivers wish to travel from *start* to *end*. There are two main paths the drivers can take. They can either travel along the path *start-A-end* or along the path *start-B-end*:



The choice is complicated by the presence of traffic. Some roads are narrow and get congested. On these roads, the travel time for every driver depends on how many travelers T also pick that path. In this network, the roads *start-A* and *B-end* are narrow and travel time is estimated to be $T/25$

minutes on average. The travel on these roads will be slower as more and more drivers choose them .

But not all roads are narrow. Some roads are so wide that they never get congested. On these roads, the travel time for every driver will be a constant number of minutes. In this network, the roads *A-end* and *start-B* are wide and travel time is estimated to be 50 minutes on average.

If every driver is optimizing travel times, as is natural in real life, how long will it take to travel from *start* to *end*?

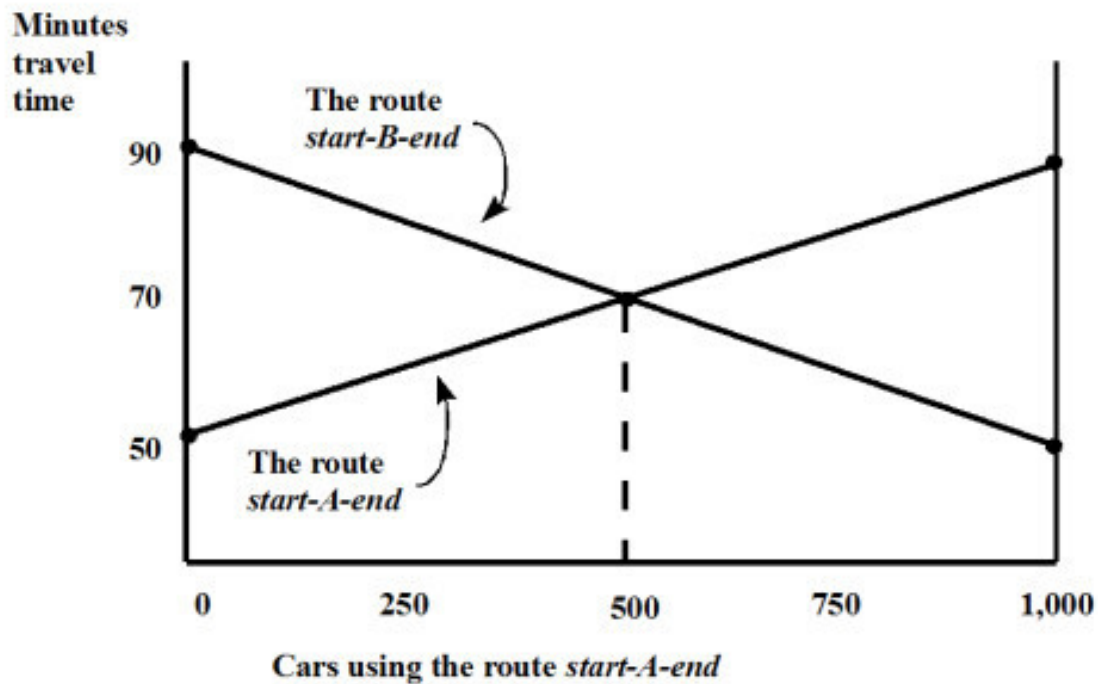
The equilibrium

The traffic game is dynamic. Each driver has to choose a path by guessing what others will do. We can start the analysis by calculating the travel times of each route dependent on the number of drivers.

If A drivers choose the route *start-A-end*, then the route will take $A / 25$ minutes for *start-A* and then 50 minutes for *A-end* .

Similarly, if B drivers choose the route *start-B-end*, then the route will take 50 minutes for *start-B* and then $B / 25$ minutes for *B-end* .

We can graph the travel times for the two paths as follows:



In practice, there are many possible outcomes to the traffic game. For instance, it is possible that 600 drivers choose the path *start-A-end* yielding a travel time of 74 minutes. That would mean the other 400 drivers who took *start-B-end* only had a 66 minute drive. But drivers are smart, and with traffic reports they can improve their choice in the future. We can imagine that some of the drivers from the *start-A-end* route would change their routes the next day.

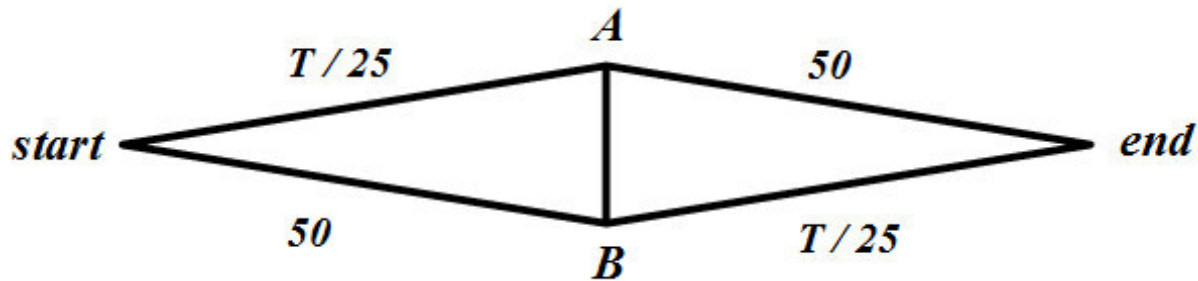
Such switching will continue as each driver optimizes. The time people stop switching—that is, when the traffic system will be in equilibrium—is therefore when the two driving routes have equal travel times.

We can solve the equations or inspect the graph to see this happens when $A = B = 500$ and both routes have a travel time of **70 minutes**. Now every driver is indifferent between the two choices and the game is in equilibrium.

What happens when you add a new road?

Imagine a new road is added between points *A* and *B*. We might imagine the road is so wide and small in length that it takes almost no time to

traverse it (it is a “free” road):



What will happen to the game now?

Solving the new game (the Braess paradox)

The new road allows drivers more choice. Now they can switch routes by going along the “free” road. How will the game play out?

The game is surprisingly simple to solve because each driver has a dominant strategy. Consider the first choice of picking the path *start-A* versus *start-B*. We can quickly see that every driver will pick *start-A*. This is because *start-A* takes 40 minutes at worst (if all 1,000 took it, it will take $1,000 / 25 = 40$ minutes) compared to *start-B* which takes 50 minutes for sure. So all drivers pick *start-A* and spend 40 minutes.

What will happen next? Once at *A*, the drivers have two choices. They can either stick to *A-end* (50 minutes for sure) or they can take the free road *A-B* (0 minutes) and follow the road *B-end* (which takes $T / 25$, and hence will be 40 minutes at worst). Naturally all drivers will choose the free road and *B-end* route.

We can conclude that all 1,000 drivers will take the path *start-A-B-end*. When we calculate the travel times, we find we have a grand total of **80 minutes** ($1,000 / 25 + 0 + 1,000 / 25$).

Additionally, notice that no driver will want to switch because the alternative routes *start-A-end* and *start-B-end* now take 90 minutes each.

In this equilibrium, the roads the travelers pick are completely congested. And consequently this game's equilibrium is 10 minutes worse than before the road *A-B* existed!

What's going on?

The Braess paradox is not just an academic curiosity. Some research suggests it might explain traffic in big cities¹ like Boston, London and New York.

The paradox is the consequence of individual incentives conflicting with the social optimum. If all drivers could agree not to take the “free” road *A-B*, then it would be possible that everyone could save 10 minutes. The problem is this proposal is not sustainable; individual drivers have an incentive to cheat and save time. Eventually the entire system breaks down when everyone cheats, making the roads congested. It just goes to show how hard rational social planning is.

Notes

1. The technical paper is from Hyejin Youn, Michael T. Gastner, and Hawoong Jeong. “Price of Anarchy in Transportation Networks.” *Physical Review Letters* (2008). <http://prl.aps.org/abstract/PRL/v101/i12/e128701> .

The paper was also summarized in *The Economist* article “Queuing Conundrums.” 11 Sept. 2008. Print Edition. Available online: http://www.economist.com/node/12202559?story_id=12202559

Conclusion

Try as we may to live rationally and logically, we are frequently placed into paradoxical situations. There are plenty of times where we can only know the logical decision in hindsight, forcing us to make choices where we need to “know without knowing.”

Just the other day I was in a dark room searching for the light switch. I eventually found it, and in the well-lit room, I discovered the switch was clearly labeled with the printed word “Light switch.” I had just faced the task of knowing without knowing; I needed to read the label to find the switch, but I could not read the switch until I found it and turned the lights on.

Gladly, not every paradox presents an insurmountable problem. In the light switch story, I found the light switch by pure luck. But other solutions can resolve the paradox as well. Perhaps the room could have had a night light as an aid, or perhaps the label could have been made with glow in the dark material.

This book presented 40 paradoxes in logic, probability, and game theory. I hope that one day if you find yourself in a dark room, unable to read a sign for the light switch, you too can appreciate the irony. While we may not be able to escape paradoxical situations, we can at least enjoy the times we are stuck in them, all the while finding practical solutions with the light of reason.

More from Presh Talwalkar

I love getting feedback. If you have a comment or suggestion, please email me presh@mindyourdecisions.com

About The Author

Presh Talwalkar studied Economics and Mathematics at Stanford University. His site *Mind Your Decisions* has blog posts and original videos about math that have been viewed millions of times.

Books By Presh Talwalkar

The Joy of Game Theory: An Introduction to Strategic Thinking .

Game Theory is the study of interactive decision-making, situations where the choice of each person influences the outcome for the group. This book is an innovative approach to game theory that explains strategic games and shows how you can make better decisions by changing the game.

Math Puzzles Volume 1: Classic Riddles And Brain Teasers In Counting, Geometry, Probability, And Game Theory . This book contains 70 interesting brain-teasers.

Math Puzzles Volume 2: More Riddles And Brain Teasers In Counting, Geometry, Probability, And Game Theory . This is a follow-up puzzle book with more delightful problems.

Math Puzzles Volume 3: Even More Riddles And Brain Teasers In Geometry, Logic, Number Theory, And Probability . This is the third in the series with 70 more problems.

But I only got the soup! This fun book discusses the mathematics of splitting the bill fairly.

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The Best Mental Math Tricks . Can you multiply 97 by 96 in your head? Or can you figure out the day of the week when you are given a date? This book is a collection of methods that will help you solve math problems in your head and make you look like a genius.

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Table of Contents

[13\)How Can a Part be Equal to the Whole?](#)

[Introduction](#)

[LOGICAL PARADOXES](#)

[Knowing Without Knowing](#)

[The Case of the Missing Dollar](#)

[Lowering Both Averages](#)

[What Exactly is a Pile of Sand?](#)

[The Missing Square](#)

[“Proofs” that \$1 = 2\$](#)

[The Least Interesting Number](#)

[Selected Self-Referential Paradoxes](#)

[The Debt that Must be Paid?](#)

[The Paradox of the Question](#)

[Russell's Set of All Sets](#)

[A Paradox of Motion](#)

[How Can a Part be Equal to the Whole?](#)

[Infinite Series Paradox](#)

[PROBABILITY PARADOXES](#)

[Birthday Paradox](#)

[False Positive Paradox](#)

[Simpson's Paradox](#)

[Bertrand's Box Paradox](#)

[Boy or Girl Paradox](#)

[The Monty Hall Problem](#)

[The Necktie Paradox](#)

[The Friendship Paradox](#)

[St. Petersburg Paradox](#)

[Wild Card Poker Paradox](#)

[Gambler's Fallacy](#)

[Surprise Quiz Paradox](#)

[Two Envelopes](#)

[Siegel's Paradox of Exchange Rates](#)

[Allais Paradox](#)

[Ellsberg Paradox](#)

[Random Chord](#)

[GAME THEORY PARADOXES](#)

[You Can't Choose Wrong Paradox](#)

[Parrondo's Paradox](#)

[Centipede Game](#)

[Hypergame Paradox](#)

[The Wallet Paradox](#)

[Condorcet Paradox](#)

[Apportionment Paradox](#)

[Forward Induction](#)

[Braess Paradox](#)

[Conclusion](#)

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